

RESEARCH ARTICLE

Evaluating Historical Outdoor Thermal Discomfort in the Eastern Mediterranean and the Middle East

Jennifer Karam¹ | Nareg Karaoghlanian¹ | Omar Hakla¹ | Louis Youssef¹ | Jad Bhamdouni¹ | Hari Prasad Dasari^{2,3} | Ibrahim Hoteit³  | Issam Lakkis¹

¹Department of Mechanical Engineering, American University of Beirut, Beirut, Lebanon | ²Climate Change Center, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia | ³Physical Sciences and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

Correspondence: Issam Lakkis (il01@aub.edu.lb)

Received: 10 July 2025 | **Revised:** 29 January 2026 | **Accepted:** 11 February 2026

Keywords: climate variability | heat stress | Middle East and Eastern Mediterranean | thermal comfort | thermal comfort indices

ABSTRACT

The Eastern Mediterranean and Middle East (EMME) region is a climate change hotspot experiencing increasing heat extremes. Understanding outdoor thermal discomfort in this region is critical for climate-resilient urban planning and public health. This study examines the spatial and temporal patterns of thermal discomfort across the EMME over the 1980–2023 period, using ERA5 reanalysis data for dry-bulb temperature, relative humidity and wind speed. Thermal discomfort is quantified using five indices: effective temperature, heat index, humidex, wet bulb globe temperature and discomfort index. All indices indicate increasing summer thermal discomfort across the region, with a pronounced shift around 1998 that coincides with a ‘mega El Niño’ event and a positive phase of the Indian Ocean Dipole. Cities in the central EMME exhibit the strongest warming, with an annual-mean increase of 0.54°C per decade—nearly three times the global mean. Warming is strongest in summer, with northern cities warming by up to 0.73°C per decade in summer, compared with 0.52°C per decade for the all-months (annual-mean) trend; southeastern cities show a weaker contrast, increasing by up to 0.42°C per decade in summer versus 0.40°C per decade in the annual-mean. Relative humidity generally declines in northern and central cities, including Tehran and Jerusalem (−1.41% and −1.07% per decade), while southeastern cities such as Riyadh show modest summer increases of 0.37% per decade. Diurnal variability analysis indicates that Abu Dhabi is the only city with a strong linear trend in the summer diurnal temperature range ($R^2 > 0.5$), whereas the summer diurnal relative humidity range exhibits more pronounced trends, notably in Jerusalem (−0.123% per year, $R^2 > 0.7$). Post-1998, summer warming intensified across the EMME (0.40°C–0.56°C per decade), with northern cities experiencing relative humidity declines of up to 0.66% per decade, highlighting escalating thermal discomfort.

1 | Introduction

Global warming has driven significant climate change in the Eastern Mediterranean and Middle East (EMME) region, leading to more frequent and severe droughts and extreme heat events (Zittis et al. 2022). Prolonged exposure to extreme heat

and arid conditions has intensified outdoor thermal discomfort. Because the EMME region spans diverse climates and terrains, the impacts of climate change on thermal comfort vary considerably across locations. In outdoor environments, thermal comfort primarily depends on meteorological variables, including dry-bulb temperature (T), relative humidity (RH) and wind

Abbreviations: DI, discomfort index; DRHR, diurnal relative humidity range; DTR, diurnal temperature range; EMME, Eastern Mediterranean and Middle East; ENSO, El Niño Southern Oscillations; ET, effective temperature; HD, humidex; HI, heat index; IOD, Indian Ocean Dipole; RH, relative humidity; T , temperature; WBGT, wet bulb globe temperature; WS, wind speed.

speed (WS) that govern heat exchange between the human body and its surroundings (Huang et al. 2020; Shimazaki et al. 2015). Variability in these conditions across the EMME is shaped by large-scale climatic patterns, including sea surface temperature anomalies, sea level pressure variations and surface zonal and meridional wind patterns across major oceanic basins (Zittis et al. 2022; Gandham et al. 2025).

Assessing long-term changes in outdoor thermal comfort is critical for developing effective adaptation strategies, public health initiatives and urban-planning decisions (Elnabawi and Hamza 2019). Thermal discomfort directly affects human well-being, productivity and the livability of urban spaces. Khan, Ismail, et al. (2025) quantify changes in outdoor labour productivity in the Arabian Peninsula over five decades (1974–2023) and report that labour productivity declined from around 94% to below 88% over the study period. As temperatures rise and extreme events become more frequent, thermal comfort assessments are increasingly used across applications. Targeted adaptation strategies can use thermal comfort assessments to identify vulnerable populations and prioritise interventions such as increased vegetation, shaded areas and water-based cooling systems to mitigate heat stress (Patel et al. 2017). Thermal comfort also plays a vital role in urban planning by supporting the development of sustainable cities that balance environmental, social and economic factors, thereby improving quality of life and long-term resilience (Panagopoulos et al. 2016). Furthermore, improving outdoor thermal comfort can reduce the prevalence of heat-related illnesses, such as heat exhaustion and heatstroke, particularly among vulnerable groups such as children, older adults and outdoor workers (Ioannou et al. 2022; Aghamolaei and Lak 2023).

At elevated temperatures, the human body sweats to promote evaporative cooling, dilates blood vessels near the skin surface to enhance heat dissipation, or reduces metabolic activity to limit internal heat production (Yang et al. 2014). In contrast, at lower temperatures, the body shivers to generate heat, constricts blood vessels to minimise heat loss, or increases metabolic heat production (Wu et al. 2021). Similarly, humidity affects physiological responses and thus thermal comfort (Huang et al. 2020): high humidity slows evaporation and reduces the effectiveness of sweating, while low humidity accelerates evaporation and enhances cooling (Kaynakli et al. 2014). Wind speed also strongly influences convective heat transfer between the skin and the environment. Higher wind speeds enhance convective heat loss, whereas lower wind speeds limit convective cooling, leading to higher skin temperatures and reduced thermal comfort (Rijal 2012). Therefore, examining changes in T , RH and WS is essential when assessing meteorological variability and its effects on thermal comfort, as these variables directly affect the body's ability to regulate its core temperature near 37°C (Gauthier, n.d.). Although global warming is commonly cited as the main driver of changes in T , RH and WS, non-anthropogenic factors also contribute to climate variability in the EMME region (Cook et al. 2015; Donat et al. 2014). These include large-scale climate modes such as the El Niño–Southern Oscillation (ENSO), Indian Ocean Dipole (IOD) and North Atlantic Oscillation (NAO), which are linked to large-scale fluctuations in oceanic and atmospheric properties. Among these, ENSO is a key driver of episodic droughts worldwide (Neelin

et al. 2003). G. Wang (2005) report a 22% global expansion of hyper-arid areas (including Middle East countries) in the 20th century, with ENSO events in the 1980s contributing substantially to this expansion. Hoell et al. (2024) also find that drought and heat over the Middle East and Southwest Asia in recent years are related to the La Niña phase of ENSO. The IOD influences regional climates around the Indian Ocean and affects global patterns via atmospheric teleconnections, motivating efforts to improve IOD prediction and assess broader impacts (Saji and Yamagata 2003; Schott et al. 2009). As a regional example, Polonsky and Basharin (2017) demonstrate that the IOD significantly affects surface air temperatures over the Mediterranean and Black Sea region during summer. Hence, establishing reliable long-term trends in thermal comfort levels—considering changes in T , RH and WS—requires accounting for the influence of ENSO and IOD events (Neelin et al. 2003; S. Ullah, Aldossary, et al. 2024).

Thermal comfort indices quantify perceived thermal conditions in outdoor environments. More than 165 indices have been proposed, each differing in input parameters, climate sensitivity and thermal-neutral range (Potchter et al. 2018; P. Kumar and Sharma 2020). de Freitas and Grigorieva (2017) review these indices and examine their adoption frequency, finding that applicability depends on comprehensiveness, scope, sophistication, transparency and validity. Published studies typically focus on a single index applied to a specific region. Dasari et al. (2021) study the variability and trends of the discomfort index in the Arabian Peninsula over a 39-year period, and Lukić (2019) uses the humidex for Belgrade, Serbia, from 1999 to 2018. Khan, Kamruzzaman, and Shahid (2025) provide an hourly assessment of the wet bulb globe temperature (WBGT) index over South Asia for 1984–2023. Other studies compare two indices, such as Santee and Wallace (2005), who evaluate the heat index and humidex to determine which better predicts heat-injury risk. Khan, Ahmad, and Shahid (2025) compare the universal thermal comfort index (UTCI) and the WBGT index in capturing extreme heat stress in major inland and coastal heatwave events affecting South Asian cities. These studies highlight the need for multi-index evaluations that compare predictions and assess sensitivity to key meteorological parameters.

Quantifying outdoor thermal comfort in the EMME region using multiple indices supports compliance with international standards that set threshold limit values for heat exposure based on different measures (Santee and Wallace 2005). To our knowledge, no study has conducted such a comprehensive evaluation. To this end, this study examines thermal comfort levels over the EMME (20–60° E, 10–50° N) across 43 years (1980–2023), focusing on the hot summer season. We analyse spatial and temporal variations in ERA5 reanalysis atmospheric data for each summer month (May–September). We assess interannual daytime and nighttime variations in T and RH to identify distinct diurnal patterns, reflecting differences in radiative processes, urban heat island effects and boundary-layer dynamics. Thermal comfort levels are evaluated using five distinct indices, examining regional variability and relationships with key atmospheric parameters. We also investigate the frequency and intensity of hours of thermal discomfort per day for major cities, and the interannual variability in the number of summer days containing thermally uncomfortable hours (as classified by the indices),

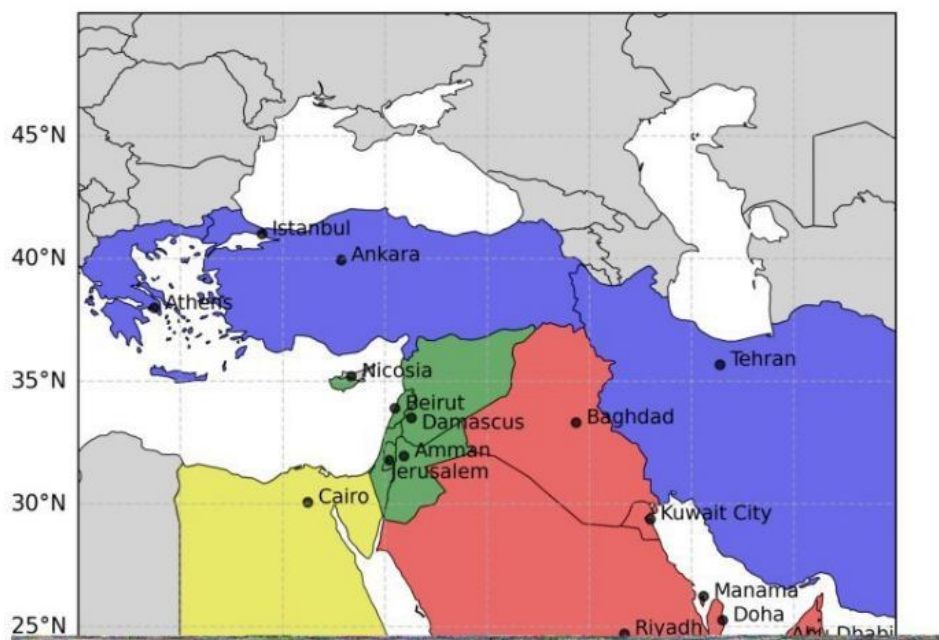


FIGURE 1 | Study region of the Eastern Mediterranean and Middle East.

enabling identification of a potential climatic shift. Finally, we relate these anomalies to ENSO phases (El Niño/La Niña) and IOD events to explore potential links between large-scale climate variability and regional heat stress. This comprehensive analysis provides essential insights into thermal discomfort in the EMME region, supporting policymakers, urban planners and researchers in developing sustainable and climate-resilient strategies.

2 | Data and Methodology

This section outlines the study region, the datasets used and the methodology adopted to compute the thermal comfort indices. It also describes the specific indices chosen for evaluation and the criteria for their selection.

2.1 | Study Region and Data Set

This study focuses on the EMME region shown in Figure 1. The domain includes Bahrain, Iraq, Saudi Arabia, Kuwait City, Qatar, the United Arab Emirates, Oman and Yemen in the southeast; Egypt in the southwest; Cyprus, Jordan, Lebanon, Palestine and Syria in the central area; and Greece, Iran and

Turkey in the north. The domain spans 20–60° E and 10–50° N (Simplemaps 2023).

The analysis is based on ERA5 hourly reanalysis data for atmospheric variables, including dry-bulb temperature (T), relative humidity (RH) and wind speed (WS), covering the period 1980–2023 (Hersbach et al. 2023). The dataset has a spatial resolution of $0.25^\circ \times 0.25^\circ$ and has been widely used in the region for thermal comfort and heat-stress assessments (Olano Pozo et al. 2024; Wong et al. 2024; Ragatoa et al. 2024) as well as for characterising spatiotemporal climatic patterns (Barbariol et al. 2021), climate extremes (Al-Sakkaf et al. 2024), cyclonic activity (Aragão and Porcù 2022) and precipitation trends (Varlas et al. 2022). The suitability of ERA5 for climate- and heat-stress-related analyses in the EMME region has been demonstrated through comparisons with ground-based observations. Validation efforts over Turkey (Yilmaz 2023), the Arabian Peninsula (W. Ullah, Alabduoli, et al. 2024) and the UAE (Nelli et al. 2024) show that ERA5 reliably captures observed seasonal variability, long-term trends and temperature extremes.

The multivariate ENSO index (MEI) (Wolter and Timlin 1998) is used to identify ENSO phases (El Niño or La Niña) over 1980–2023. The MEI incorporates six variables to represent ENSO dynamics: sea surface temperature, sea level pressure,

TABLE 1 | Thermal comfort indices and corresponding temperature ranges. [Colour table can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Index	Input variables	Equation (Supporting Information)	Temperature range (°C)	References
Effective temperature index (ET)	T, RH and WS	Equation (S.1)	1 to 45	Missenard (1933)
Heat index (HI)	T and RH	Equation (S.2)	20 to 60	Steadman (1979)
Humidex (HD)	T and RH	Equation (S.3)	21 to 49	Lally and Watson (1960)
Wet bulb globe temperature (WBGT)	T and RH	Equation (S.5)	10 to 50	Yaglou and Minaed (1957)
Discomfort index (DI)	T and RH	Equation (S.7)	20 to 40	Thom (1959)
Universal thermal comfort index (UTCI)	T, RH, WS and radiation fluxes	Equation (S.8)	−50 to +50	Jendritzky et al. (2012)

Note: The range of RH is 0%–100%.

zonal and meridional surface winds, air temperature and cloud cover over the tropical Pacific Ocean. Positive and negative IOD events are identified using the Dipole Mode Index (DMI), defined as the difference in anomalous sea surface temperature between the western equatorial Indian Ocean and the south-eastern equatorial Indian Ocean (Saji and Yamagata 2003). MEI and DMI data are obtained from the National Oceanic and Atmospheric Administration (NOAA) (CPC, National Oceanic and Atmospheric Administration 2024).

2.2 | Thermal Comfort Indices

Outdoor thermal comfort indices are generally classified into three categories (Coccolo et al. 2016). The first category is based on human energy balance, which links body physiology to environmental parameters. The second category consists of empirical indices, typically expressed as linear regressions derived from field studies. The third category comprises weather-based indices that quantify thermal comfort as a function of the thermal environment, typically without explicitly accounting for microclimate effects or human behaviour. These indices are typically expressed as functions of meteorological variables such as WS, net radiation, T , RH and water-vapour pressure.

In this study, we analyse five commonly used thermal comfort indices: effective temperature (ET), heat index (HI), humidex (HD), WBGT and discomfort index (DI) (also referred to as the temperature-humidity index, THI). These indices depend on commonly measured atmospheric conditions (T , RH and, where applicable, WS) and generally correspond to typical climatic conditions in the EMME region, particularly during hot summer months; however, during cooler periods, some values may fall outside their recommended validity ranges (Table 1). In addition, we include the UTCI to provide a physiologically based assessment of thermal sensation in the EMME region and to facilitate comparison with the indices that rely solely on meteorological variables. The UTCI values are obtained from ERA5-HEAT at hourly temporal resolution and a spatial resolution of $0.25^\circ \times 0.25^\circ$. These values are computed using an operational approximation to the full UTCI-Fiala thermoregulation model, incorporating a parameterized offset function.

Each index value is mapped to categorical ranges (or zones) that reflect thermal sensation and comfort (Coccolo et al. 2016). Table 2 summarises the thermal sensation scales and applicable ranges for each index, spanning conditions from very cold to sweltering. Detailed definitions and formulas for the indices, expressed in terms of the relevant meteorological variables, are provided in the Supporting Information.

3 | Results

In what follows, we examine the spatiotemporal distribution of the considered meteorological variables (T , RH and WS) and thermal comfort indices (ET, HI, HD, WBGT and DI) on seasonal and interannual timescales for major cities in the EMME region over the 43-year study period.

3.1 | Analysis of Meteorological Variables

Figure 2 shows linear-trend maps of decadal rates of change in T and RH across the EMME region for 1980–2023. Except for a few localised areas, most of the domain exhibits a clear warming trend. Table 3 summarises annual-mean and summer decadal changes in T and RH for the central, southwestern, southeastern and northern city groups shown in Figure 1.

Central cities exhibit the largest annual-mean temperature increase of 0.54°C per decade, nearly three times the global average of around 0.2°C per decade since the late 1970s (Copernicus Climate Change Service 2025). Comparing annual-mean and summer trends shows that warming is most pronounced in summer (May–September). For northern cities, annual-mean T increases by 0.52°C per decade, with a summer increase of 0.73°C per decade. For southeastern cities, annual-mean T increases by 0.40°C per decade, with a summer increase of 0.42°C per decade. The increase in T is accompanied by an annual reduction in RH; however, in summer, southeastern and southwestern cities experience increases of 0.33% and 0.10% per decade, respectively.

We focus on summer because it represents the peak season for thermal stress in the EMME region. Summer is characterised

TABLE 2 | Thermal sensation and thermal comfort scale of each thermal comfort index. [Colour table can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Thermal sensation scale	Index						Thermal comfort scale
	ET	HI	HD	WBGT	DI	UTCI	
Very cold	<1					< -27	
Cold	1 to 9					-13 to -27	
Cool (moderate hazard)	9 to 17					-1 to 9	
Neutral (no danger)	17 to 21		< 30	< 18	< 21	9 to 26	No discomfort
Warm (caution)	21 to 23	27 to 32	30 to 40	18 to 24	21 to 24	26 to 32	Under 50% population feels discomfort
					24 to 27		Over 50% population feels discomfort
Hot (extreme caution)	23 to 27	32 to 41	40 to 45	24 to 28	27 to 29	32 to 38	Most of population suffers discomfort
Very hot (danger)	> 27	41 to 54	45 to 55	28 to 30	29 to 32	38 to 46	Everyone feels severe discomfort
Sweltering (extreme danger)		> 54	> 55	> 30	> 32	> 46	State of medical emergency

by extreme temperatures and elevated outdoor heat exposure, making it the most relevant period for evaluating outdoor thermal comfort and informing climate-adaptation and resilience strategies.

Figures 3–5 show the spatial distributions of T , RH and WS over the study region, averaged for each summer month (May–September) over the 43-year study period. City values reported below represent 43-year summer means. The highest temperatures occur in July and August, with maxima primarily over southeastern regions and, to a lesser extent, the southwest. These regions include Kuwait City (35.81°C), Baghdad (35.26°C), Riyadh (34.47°C), Manama (32.08°C), Doha (33.96°C), Abu Dhabi (32.45°C), Muscat (33.26°C), Sana'a (20.65°C) and Cairo (27.66°C). Lower temperatures are observed in central cities such as Nicosia (27.24°C), Beirut (24.48°C), Amman (24.36°C), Jerusalem (25.40°C) and Damascus (23.78°C), followed by northern cities including Istanbul (21.32°C), Tehran (24.10°C) and Athens (24.75°C).

Relative humidity exhibits contrasting spatial patterns. The driest regions are located in the southeast, notably Kuwait City (15.30%), Riyadh (12.96%) and Baghdad (18.93%). In contrast, higher RH values are recorded in central areas, including Beirut (67.37%), Damascus (39.04%) and Jerusalem (51.66%). High RH is also observed near large water bodies, particularly the Black and Mediterranean Seas. The highest RH levels (70%–85%) occur in cities near the Black Sea, such as Istanbul (72.92%). Along the Mediterranean coast, RH values range from 55% to 70%, gradually decreasing towards inland southeastern regions to as low as 10%–40%.

For WS, the highest values are observed over desert regions, including Baghdad (3.97 m s⁻¹), Kuwait City (5.44 m s⁻¹), Manama (4.61 m s⁻¹), Doha (4.32 m s⁻¹) and Abu Dhabi (3.84 m s⁻¹). These winds largely arise from Shamal flow, characterised by north-westerly winds (Hamzeh et al. 2021).

Decadal summer changes in T and RH for major cities provide city-specific insights into long-term climate trends and their potential urban impacts. Seasonal and annual averages per decade of T and RH for major cities in the southern, central and northern parts of the study region are listed in Tables S.1, S.2 and S.4 of the Supporting Information, respectively. Table 4 lists the cities with the largest average increase in summer T per decade, along with the corresponding RH change. The results show that warming is strongest in summer (May–September), particularly in northern cities such as Athens and Tehran, where T increases by 1.13°C and 0.69°C per decade, respectively. Central cities also exhibit notable summer warming, including Jerusalem (0.68°C per decade) and Amman (0.64°C per decade). Relative humidity shows a declining trend in most regions. The fastest summer decreases occur in northern and central cities, especially Tehran, Jerusalem and Amman, where RH decreases by 1.41%, 1.07% and 0.92% per decade, respectively. Conversely, summer RH increases in southeastern cities such as Riyadh (0.37% per decade) and Baghdad (1.07% per decade). Overall, T increases across all regions, whereas RH decreases in most regions. Rising T and decreasing RH can have opposing effects on thermal comfort: higher T increases discomfort, whereas lower RH can partially offset it. This interplay is illustrated in Figure 7.

To examine differences between daytime and nighttime variations in T and RH, we computed annual summer means and used them to estimate the diurnal temperature range (DTR) and diurnal relative humidity range (DRHR) over the summer season. Table 5 summarises linear trends in DTR and DRHR for various cities, sorted by the R^2 value of the linear fit. Abu Dhabi shows the most pronounced change in summer DTR, with its annual trends illustrated in the bottom row of Figure 6. The top row shows that daytime and nighttime trends differ in slope for both T and RH, with the absolute difference between day and night values narrowing over time. Among all cities analysed, Abu Dhabi is the only one with $R^2 > 0.5$ for the summer DTR

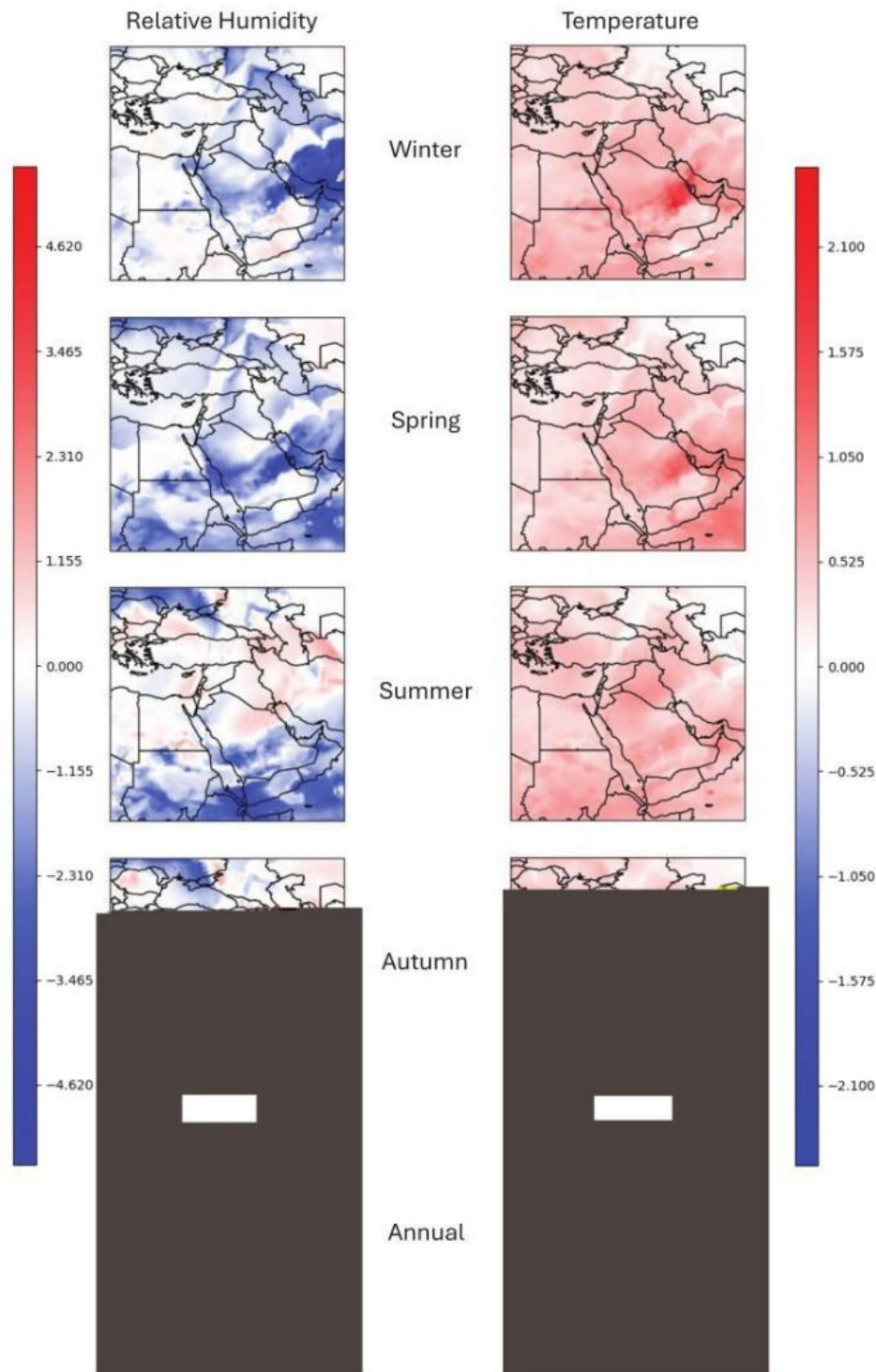


FIGURE 2 | Regression maps showing the average change in relative humidity % (left) and dry-bulb temperature °C (right) per decade.

trend. In contrast, summer DRHR trends are more pronounced across multiple cities. For instance, Jerusalem exhibits a decrease in summer DRHR at a rate of 0.1229% per year, with a strong linear fit ($R^2 > 0.7$).

A declining summer DTR indicates that nighttime temperatures increase more rapidly than daytime temperatures, consistent with increasingly oppressive nocturnal heat conditions. Such behaviour is consistent with broader warming and may

be enhanced by urbanisation and radiative effects that reduce nocturnal cooling. The long-term decline in both daytime and nighttime summer RH, with a steeper decrease at night, suggests an overall drying tendency in the region. The shrinking DRHR reflects reduced nocturnal moisture recovery, which may be related to warmer nights, reduced surface moisture availability and/or changes in boundary-layer stability. This pattern may amplify surface aridity and thermal stress, particularly during extended dry seasons.

TABLE 3 | Annual and summer decadal variations of T and RH and their average summer values for the central, southeastern, southwestern and northern cities.

	T			RH		
	Annual decadal variation	Summer decadal variation	Summer average	Annual decadal variation	Summer decadal variation	Summer average
Central	+0.54°C	+0.58°C	26.95°C	−1.25%	−0.43%	48.18%
Southeastern	+0.40°C	+0.42°C	31.76°C	−0.21%	+0.33%	45.34%
Southwestern	+0.58°C	+0.57°C	27.81°C	−0.70%	+0.10%	52.07%
Northern	+0.52°C	+0.73°C	22.25°C	−0.55%	−0.60%	50.96%

3.2 | Comparison of the Comfort Indices in Terms of Their Dependence on Weather Parameters

In this section, we compare how different thermal comfort indices depend on key meteorological variables (T , RH and WS). This comparison helps interpret similarities and differences in the spatial distributions and long-term trends of the indices in terms of local atmospheric conditions. Each thermal index I is first normalised as follows:

$$I^* = \frac{\min \{ \max (I, I_L), I_H \} - I_L}{I_H - I_L} \quad (1)$$

Here, I_L and I_H denote the lower and upper bounds of the applicability range (Table 2). Contours of the normalised indices are shown in Figure 7 over the applicable ranges of T and RH. To facilitate comparison, all panels share the same temperature axis and an RH range of 0%–100%. Because ET depends on WS in addition to T and RH, Figure 7 also shows T –RH maps of ET for wind-speed values ranging from 0 to 7 m s^{−1}. For RH values greater than 20%, all indices are more sensitive to T than to RH, as indicated by the steep contour slopes. The temperature range covered by each index reflects the climatic conditions for which it was developed. For example, ET extends to low temperatures (down to 0°C), whereas HI is intended for hot conditions, with an applicability range from 20°C to 60°C.

Inspection of Figure 7 shows that DI, HI, HD and WBGT exhibit similar RH-dependence patterns within their applicable T ranges. In particular, most indices show a stronger dependence on T at high-RH values. For example, at RH = 100%, as T increases from 20°C to 40°C, DI increases from 20°C (neutral) to 40°C (sweltering), HI increases from 15.13°C (neutral) to 107.87°C (sweltering), HD increases from 27.41°C (neutral) to 75.28°C (sweltering) and WBGT increases from 24.45°C (warm) to 55.51°C (sweltering). At lower RH values, the increase with T is smaller. For example, at RH = 20%, as T increases from 20°C to 40°C, DI increases from 17.58°C (neutral) to 28.78°C (hot), HI increases from 23.20°C (neutral) to 39.38°C (hot), HD increases from 17.04°C (neutral) to 42.61°C (hot) and WBGT increases from 17.11°C (neutral) to 32.4°C (sweltering).

Compared to the other indices, ET exhibits a weaker dependence on RH, especially at higher WS. As shown in Figure 7, ET contours are nearly vertical over temperature bands that shift to higher values as WS increases (approximately 11°C at 0 m s^{−1} to 18°C at 7 m s^{−1}), indicating a minimal influence of RH in these ranges.

In contrast to the other indices, for a given WS, as RH increases from 0% to 100%, ET decreases at low T and increases at high T . For example, at WS = 3 m s^{−1}, as RH increases from 0% to 100% (relative to its value at RH = 0%), ET decreases by 36.1% at T = 12°C and increases by 35.4% at T = 30°C. High RH can influence thermal sensation differently depending on temperature. At lower temperatures, increased humidity can sometimes be associated with a cooler perceived sensation (lower ET), for example by enhancing conductive heat loss and reducing clothing insulation, while still limiting evaporative heat loss (Du et al. 2019). Conversely, at higher temperatures, increased humidity is commonly associated with a warmer, more oppressive sensation (higher ET), primarily because it reduces the evaporative cooling potential of sweating and can increase heat storage and discomfort.

The ET contours in Figure 7 also show that, as WS increases, colder sensations (lower ET) can occur at higher air temperatures within the cool-temperature range, consistent with enhanced heat exchange. For example, at WS = 1 m s^{−1} and T = 15°C, ET increases by 7% (from 10.6°C to 11.3°C) as RH increases from 0% to 100%. At WS = 3 m s^{−1}, the trend reverses and ET decreases by 4.6% (from 7.7°C to 7.3°C). The magnitude of this decrease increases to 15% (from 6.3°C to 5.4°C) at WS = 5 m s^{−1} and further increases to 25% (from 5.4°C to 4.1°C) at WS = 7 m s^{−1}.

3.3 | Thermal Comfort Analysis

We compute thermal comfort indices (ET, HI, HD, WBGT, DI and UTCI) for each summer month and present monthly mean maps (Figures 8–13), averaged over the study period (1980–2023). Colours correspond to the thermal sensation classes indicated in the legends. Grey regions denote areas where the temperature values fall outside the validity range of the respective indices (Table 1). We also compute, for each city, the mean number of discomfort hours per day for each summer month, as

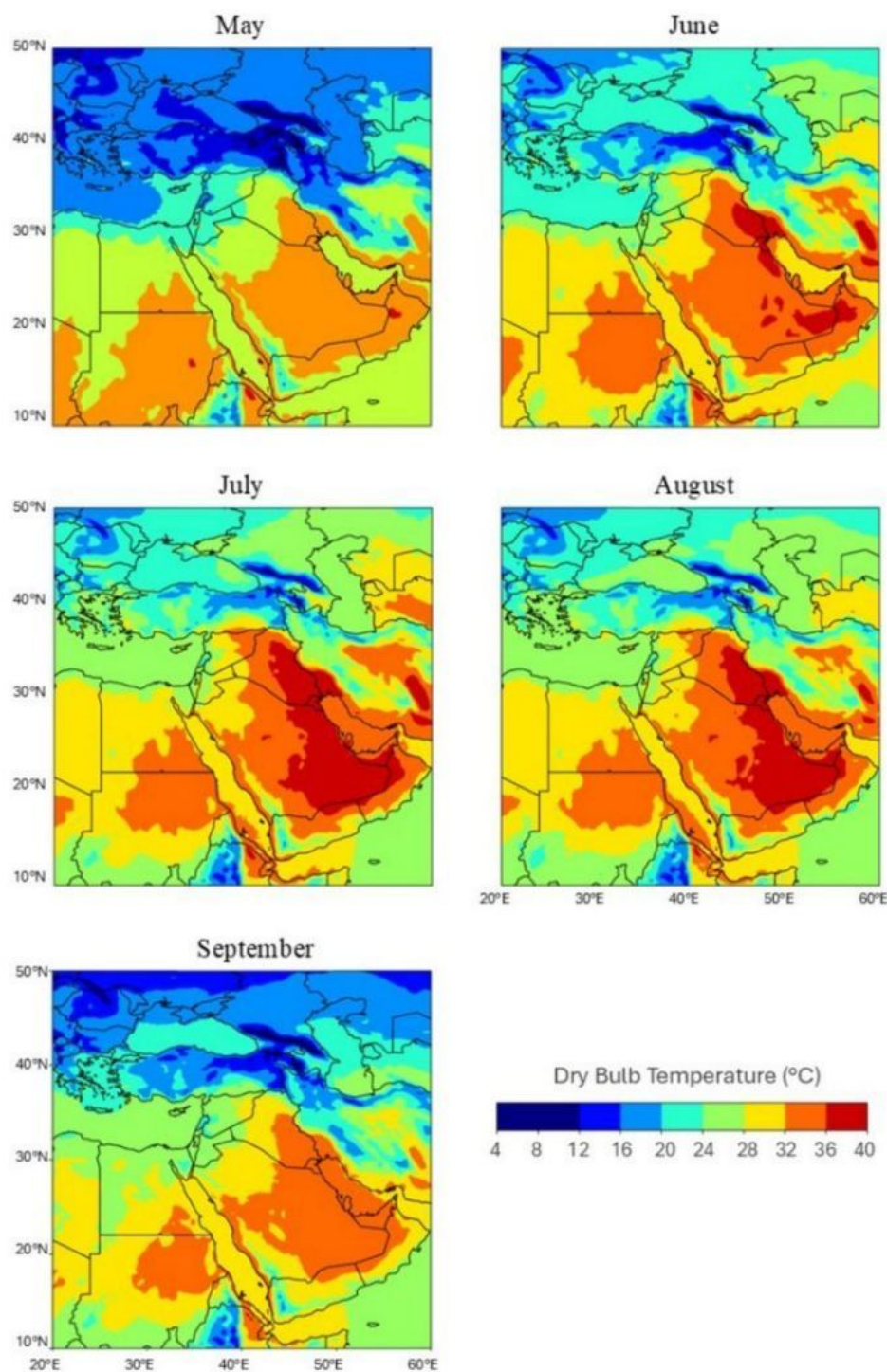


FIGURE 3 | Maps showing the monthly average dry-bulb temperature for the study region over the 43-year period.

well as standardised anomalies of the mean number of summer discomfort days.

Figure 8 shows the spatial distribution of thermal sensation based on ET for the summer months, averaged over the study period. July and August are the most extreme months in terms of thermal discomfort. Among the indices considered, ET is the only one that accounts for both cold and hot conditions. Most northern cities fall within the cold and cool categories during

May, June and September, while neutral sensations dominate in July and August, especially in Athens, Ankara and Istanbul. In the southeastern and southwestern regions, including Cairo, Baghdad and Riyadh, populations experience uncomfortable warm or hot sensations throughout the summer. Very hot conditions prevail in Muscat, Abu Dhabi, Doha, Manama and Kuwait City during July and August. The only city in the southeastern region that maintains neutral conditions throughout the summer is Sana'a. In the central region, cool sensations prevail in

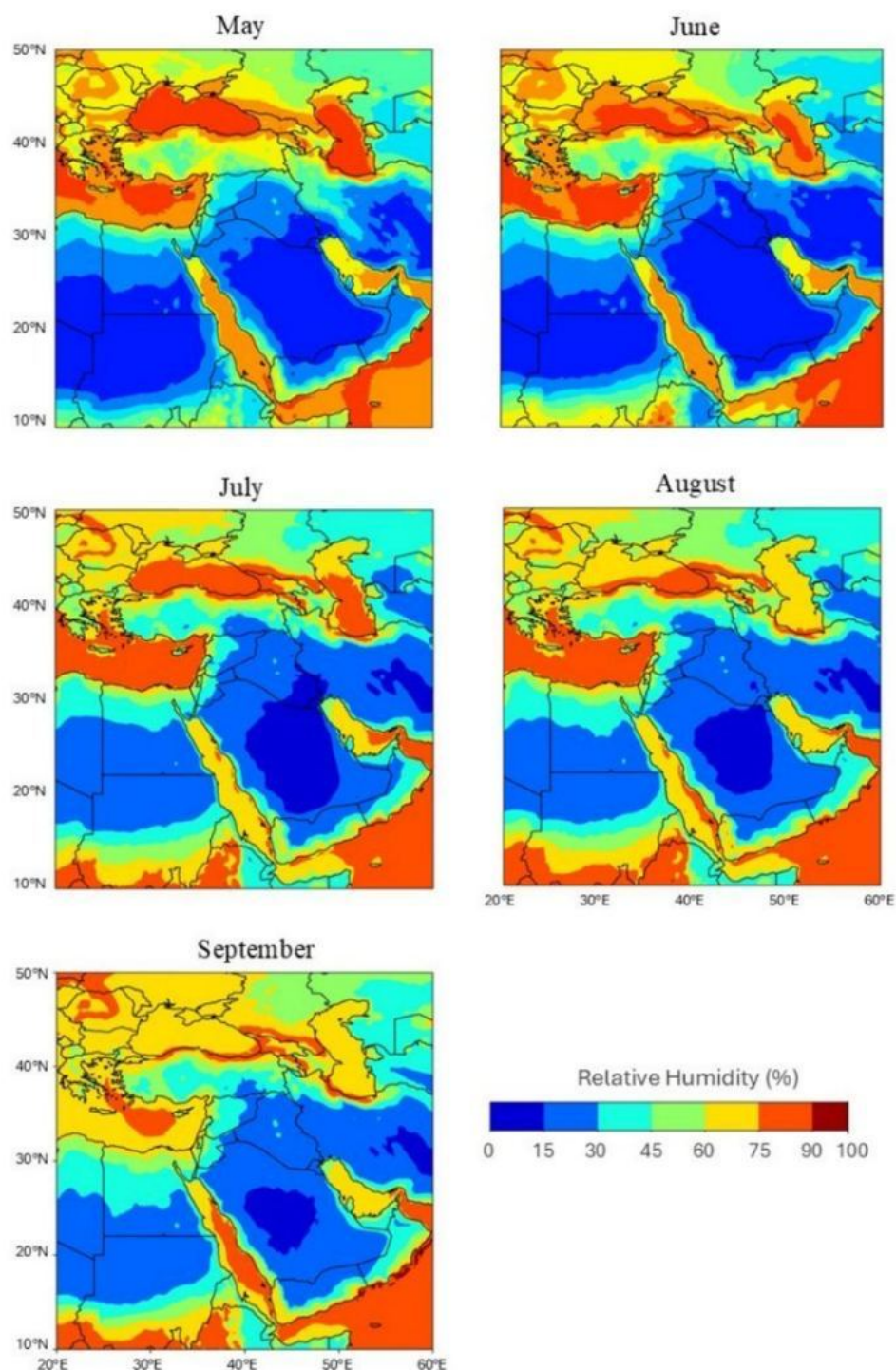


FIGURE 4 | Maps showing the monthly average RH over the study region for the 43-year period.

May; neutral sensations are perceived in June and September; and warm sensations dominate in July and August, particularly in Damascus and Jerusalem.

Similar spatial patterns are observed for HI (Figure 9). However, in most northern areas, HI values fall out of range because the index is designed for hot climatic conditions. As with ET, July and August are the most extreme months. During these months, central cities such as Beirut and Jerusalem experience neutral sensations, while inland cities such as Amman and Damascus

experience warm sensations. Hot sensations are observed in Riyadh, Baghdad, Kuwait City, Manama, Doha, Muscat and Abu Dhabi. Cairo exhibits warm conditions during the peak summer months. Sana'a remains out of range throughout the summer. Notably, ET is highly dependent on RH only at high T (Figure 7); in comparison, HI predicts lower discomfort in the hot southeastern cities.

The summer-average distribution of HD (Figure 10) shows the lowest overall discomfort levels among all indices. This

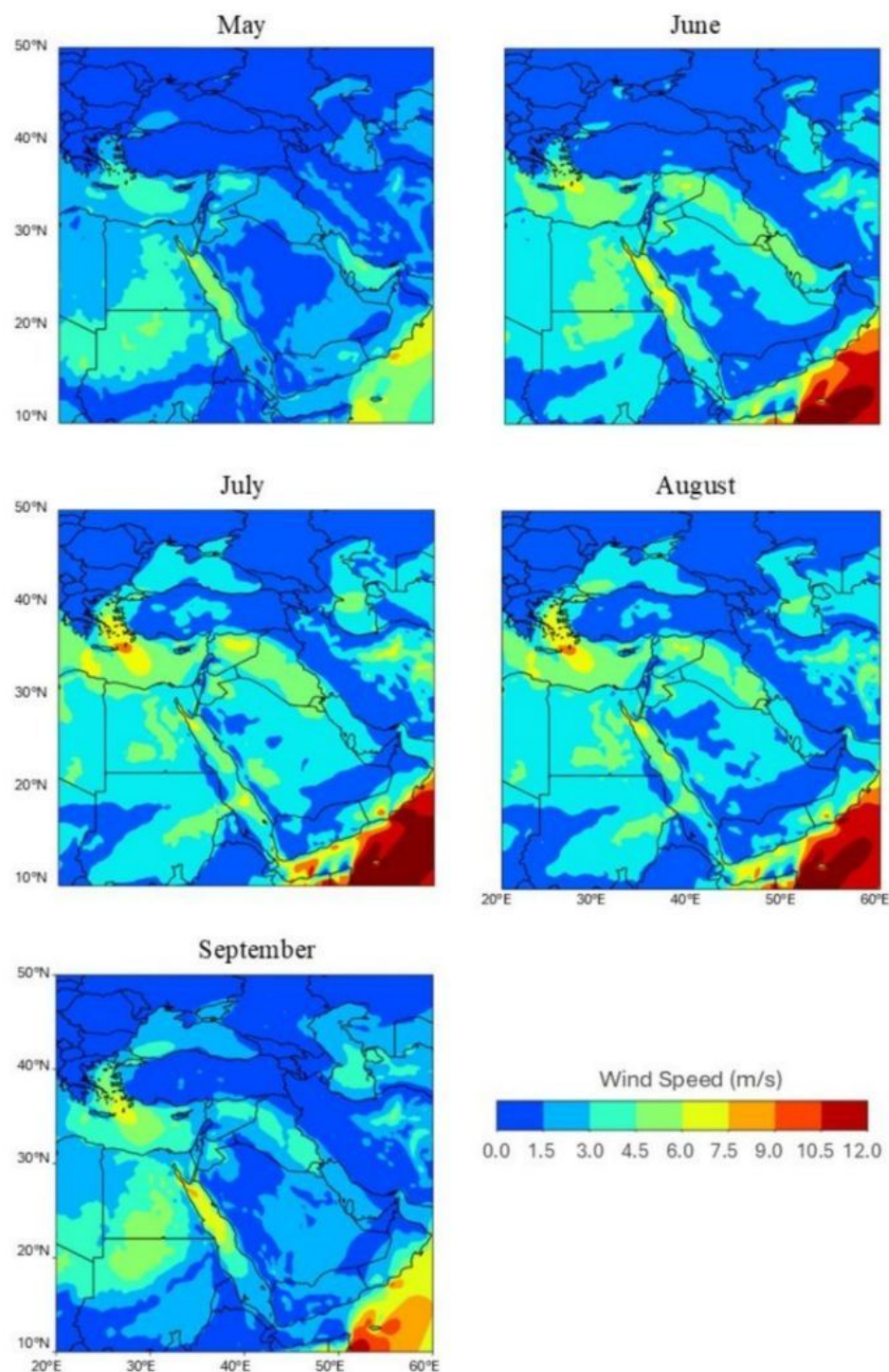


FIGURE 5 | Maps showing the monthly average WS over the study region for the 43-year period.

behaviour is attributed to its strong dependence on RH (Figure 7). As shown in Figures 3 and 4, the regions with the highest T also exhibit the lowest RH, resulting in lower HD values. In northern regions, HD generally falls out of range in May, June and September. Neutral conditions are identified in Tehran, Istanbul and Ankara, while warm sensations are observed in Athens during July and August. Compared to HI, HD classifies Beirut as out of range in May and neutral from June through September. Cairo is classified as neutral in May and warm in the subsequent summer months. In southeastern

regions, HD predicts warm conditions in Baghdad, Riyadh and Kuwait City, and hot conditions in Muscat, Abu Dhabi, Manama and Doha. Sana'a remains neutral throughout the summer, consistent with ET.

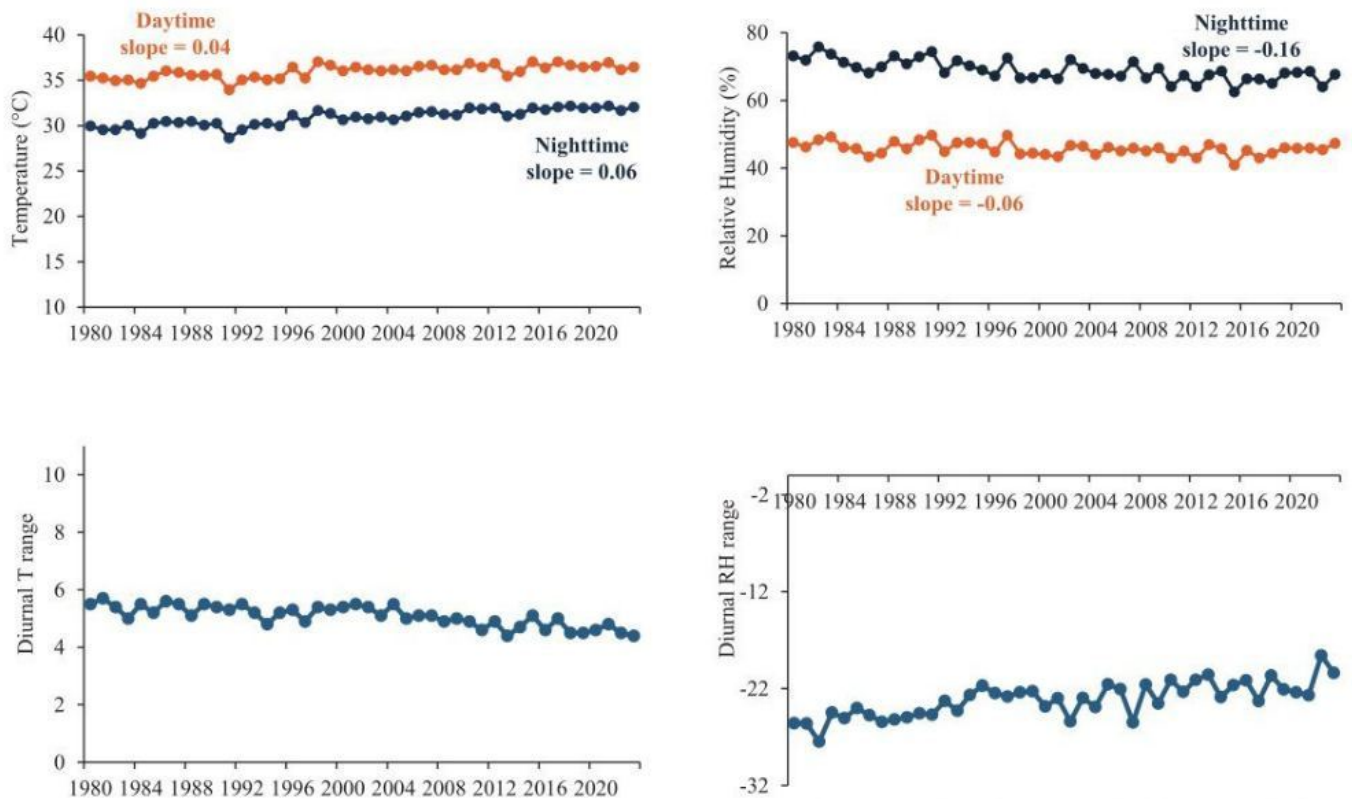
The WBGT results shown in Figure 11 exhibit the highest levels of discomfort among all indices. The southeastern region experiences persistent hot to very hot thermal sensations throughout the summer, with sweltering sensations in Kuwait City, Manama, Abu Dhabi, Muscat and Doha during

TABLE 4 | Top cities witnessing the highest variations of decadal summer T and RH.

Cities	T		RH	
	Summer decadal variation	Summer average	Summer decadal variation	Summer average
Athens	+1.13°C	24.90°C	−0.58%	51.98%
Tehran	+0.69°C	22.80°C	−1.41%	28.76%
Riyadh	+0.69°C	34.63°C	+0.37%	14.12%
Jerusalem	+0.68°C	25.56°C	−1.07%	54.96%
Amman	+0.64°C	24.52°C	−0.92%	48.52%
Baghdad	+0.46°C	35.66°C	+1.07%	19.58%
Abu Dhabi	+0.35°C	33.58°C	−0.72%	57.33%

TABLE 5 | Left: Averaged summer change in $T_{\text{day}} - T_{\text{night}}$ per year, right: Averaged summer change in $\text{RH}_{\text{day}} - \text{RH}_{\text{night}}$ per year.

T			RH		
City	Slope	R^2	City	Slope	R^2
Abu Dhabi	−0.0222	0.617	Jerusalem	0.123	0.713
Doha	−0.0187	0.487	Cairo	0.102	0.634
Nicosia	−0.0167	0.479	Amman	0.107	0.623
Istanbul	0.00999	0.400	Beirut	0.0692	0.594
Beirut	−0.00817	0.380	Nicosia	0.106	0.580
Cairo	−0.0126	0.348	Abu Dhabi	0.106	0.577

**FIGURE 6** | Top: Interannual variations in summer dry-bulb temperature (left) and relative humidity (right) during daytime (orange line) and nighttime (black line) for the 43-year period in Abu Dhabi. Bottom: Difference between daytime and nighttime summer averages.

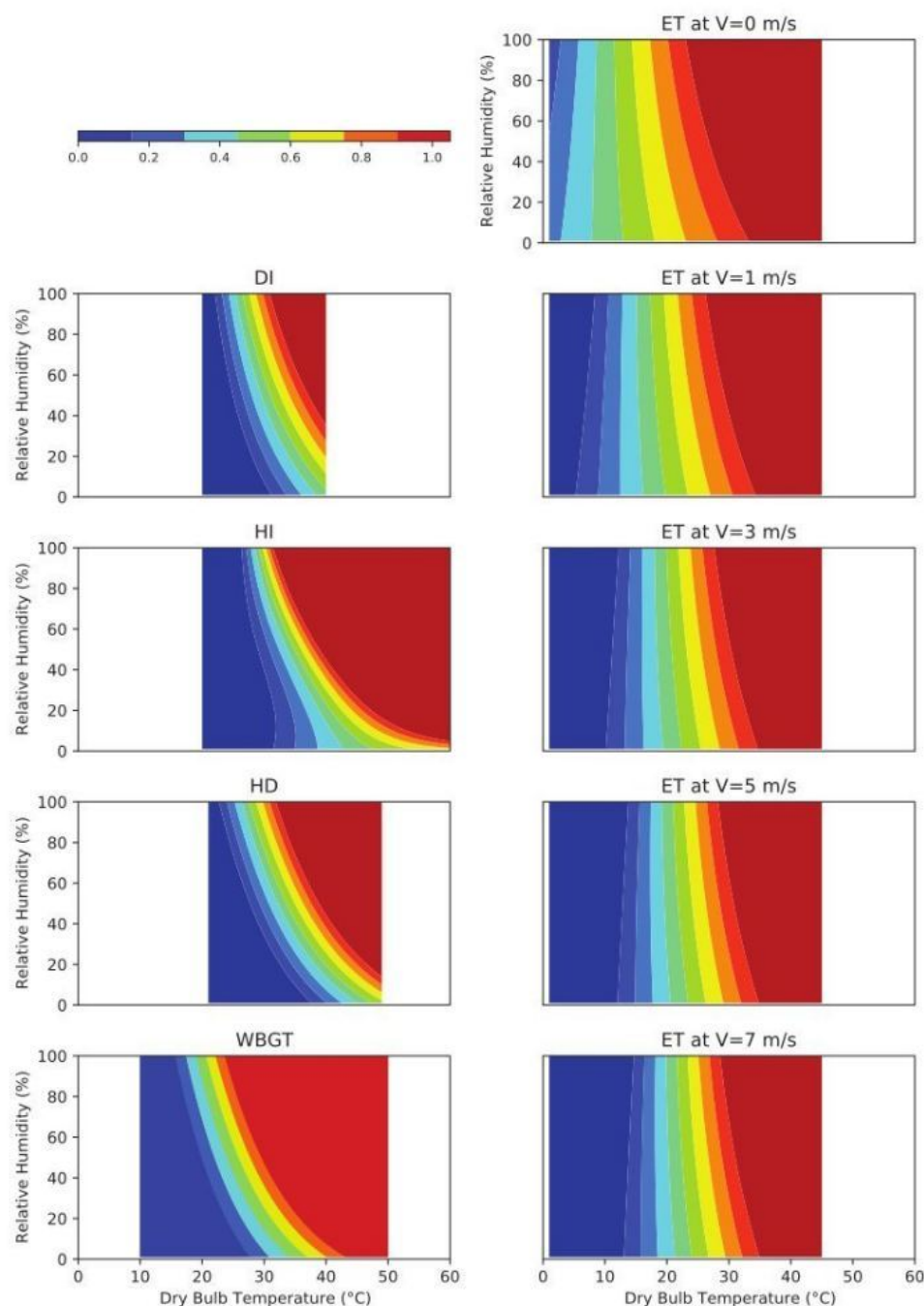


FIGURE 7 | Colour maps showing (left) the normalised thermal comfort indices as functions of air temperature and RH, and (right) the normalised ET as a function of air temperature and RH for different wind speed values.

June, July and August. Sana'a experiences warm sensations throughout the summer, while Cairo experiences warm conditions in May and hot sensations from June through September. In central cities, warm sensations are observed in May, June and September, while hot to very hot conditions dominate in July and August. Due to its strong dependence on T and weak sensitivity to RH (Figure 7), the WBGT maps closely resemble those of T (Figure 3).

The DI results shown in Figure 12 resemble those of HD, as both indices exhibit a strong dependence on RH (Figure 7). DI tends to underestimate severe heat stress in hot and arid regions, particularly in southeastern cities (Figures 3 and 4). In northern cities,

DI falls out of range in May and for some locations, September, while moderately warm sensations are predicted for Athens in June–September. Cairo and the southeastern cities exhibit warm sensations throughout summer, except for Kuwait City, Doha, Baghdad, Manama, Abu Dhabi and Muscat, which reach hot sensations during July and August. Sana'a remains neutral, consistent with the predictions of HD and ET. The central region is generally classified as moderately warm, with localised neutral conditions in May.

Figure 13 presents the spatial distribution of thermal sensation derived from UTCI, which explicitly incorporates human physiological parameters. The resulting patterns closely

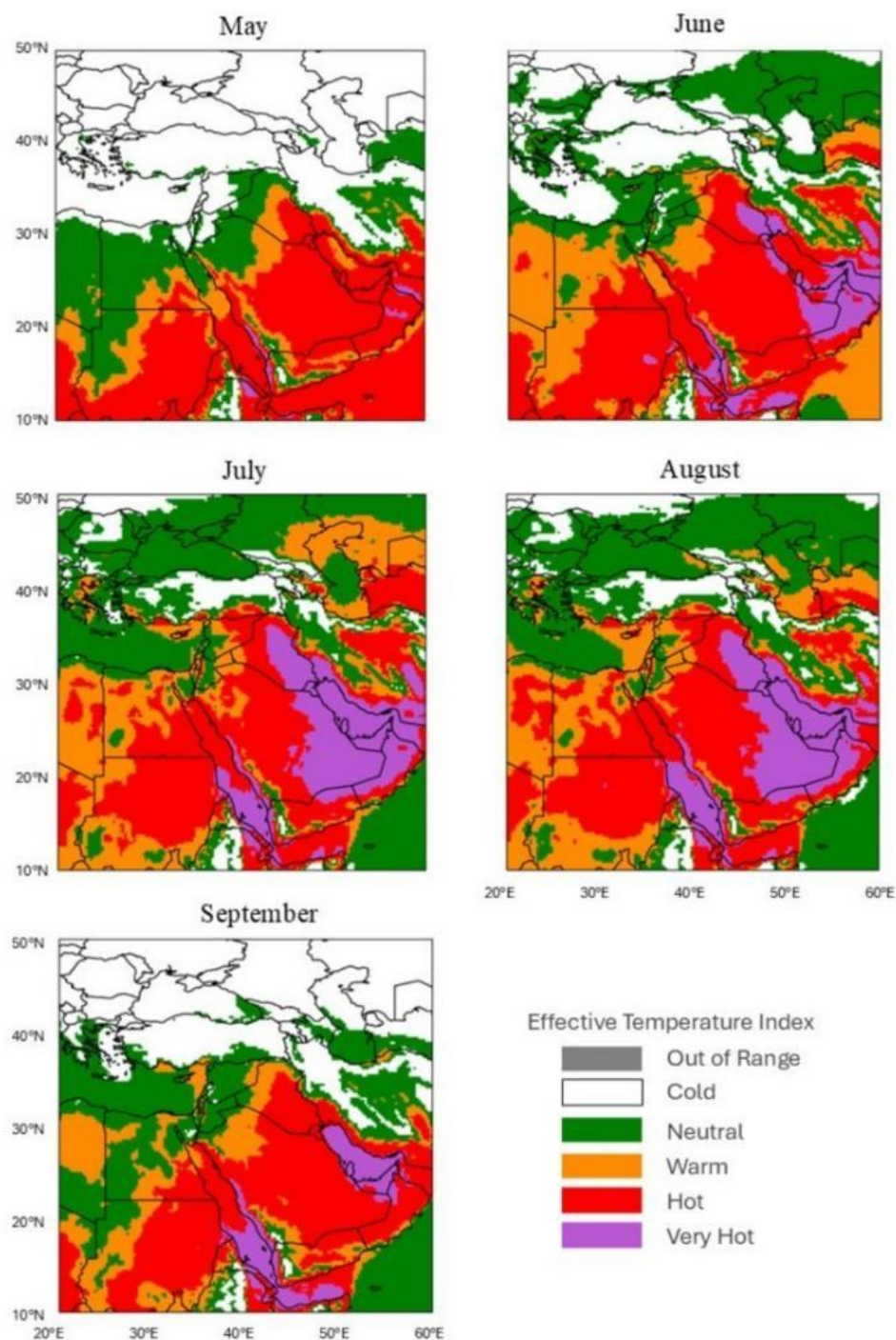


FIGURE 8 | Colour maps showing the 1980–2023 average of the ET index in the study region for May to September.

resemble those of HI (Figure 9). Both indices indicate warm and hot sensations in the southwestern and southeastern regions, respectively, during June, July and August. In May and September, both indices predict warm conditions in the southwestern region, with hot conditions confined to its eastern corner. In contrast, in the northern region and parts of the central region, HI values fall out of range, while UTCI indicates neutral thermal sensation for all summer months. This discrepancy arises from the absence of a neutral class in the HI formulation (Table 2) and does not imply a physical contradiction. A further minor discrepancy appears in the

lower-right corner of the domain, where the two indices differ by one thermal sensation class: HI indicates hot while UTCI indicates warm during May and June, and HI indicates warm while UTCI indicates neutral during the remaining summer months. This difference is negligible, as it occurs over a water body where human thermal sensation is of low practical relevance due to sparse human presence (primarily limited to marine traffic).

Overall, incorporating human physiological parameters through UTCI does not substantially alter the spatial patterns of thermal

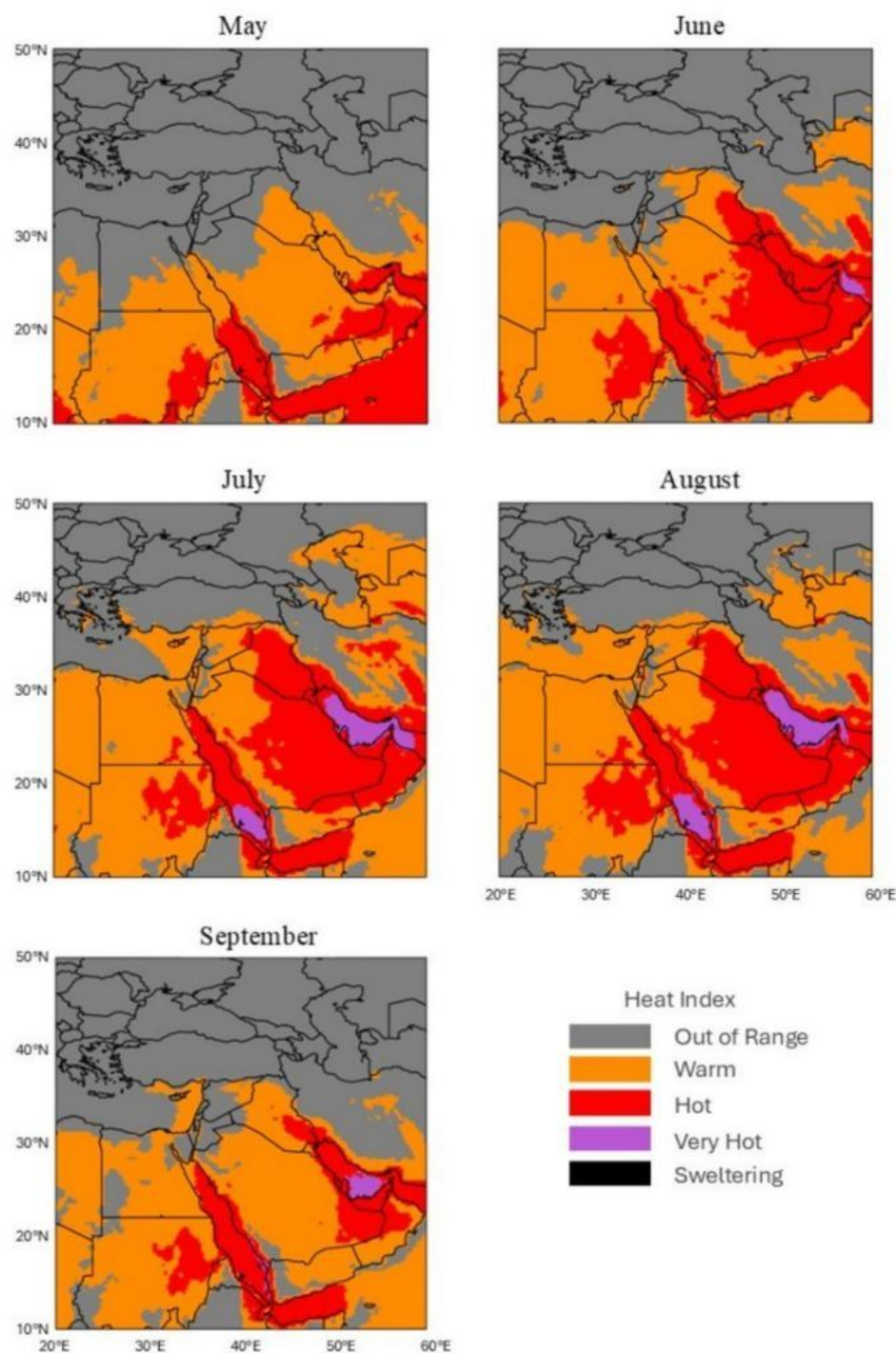


FIGURE 9 | Colour maps showing the 1980–2023 average of HI in the study region for May to September.

sensation relative to indices based solely on meteorological variables. Given the close agreement between the patterns predicted by UTCI and those derived from HI, additional analysis of UTCI is not pursued in this study.

To identify regions of unambiguous heat stress, we mapped zones where all indices consistently indicate the same thermal sensation category. This agreement enhances confidence in the results and supports policy applications, as different sectors (e.g., field operations, agriculture and urban planning) tend to

favour different indices depending on their needs. Figure 14 illustrates the spatial zones across the EMME region where the five indices (ET, HI, HD, WBGT and DI) converge on the same thermal sensation category for each summer month (May–September), averaged over 1980–2023. The indices consistently classify the southeastern part of the region, including cities such as Riyadh, Baghdad and Kuwait City, as experiencing warm to hot thermal sensations throughout the summer. Very hot to sweltering conditions are identified in Doha, Manama and Abu Dhabi, while Cairo is classified as warm to hot during July and

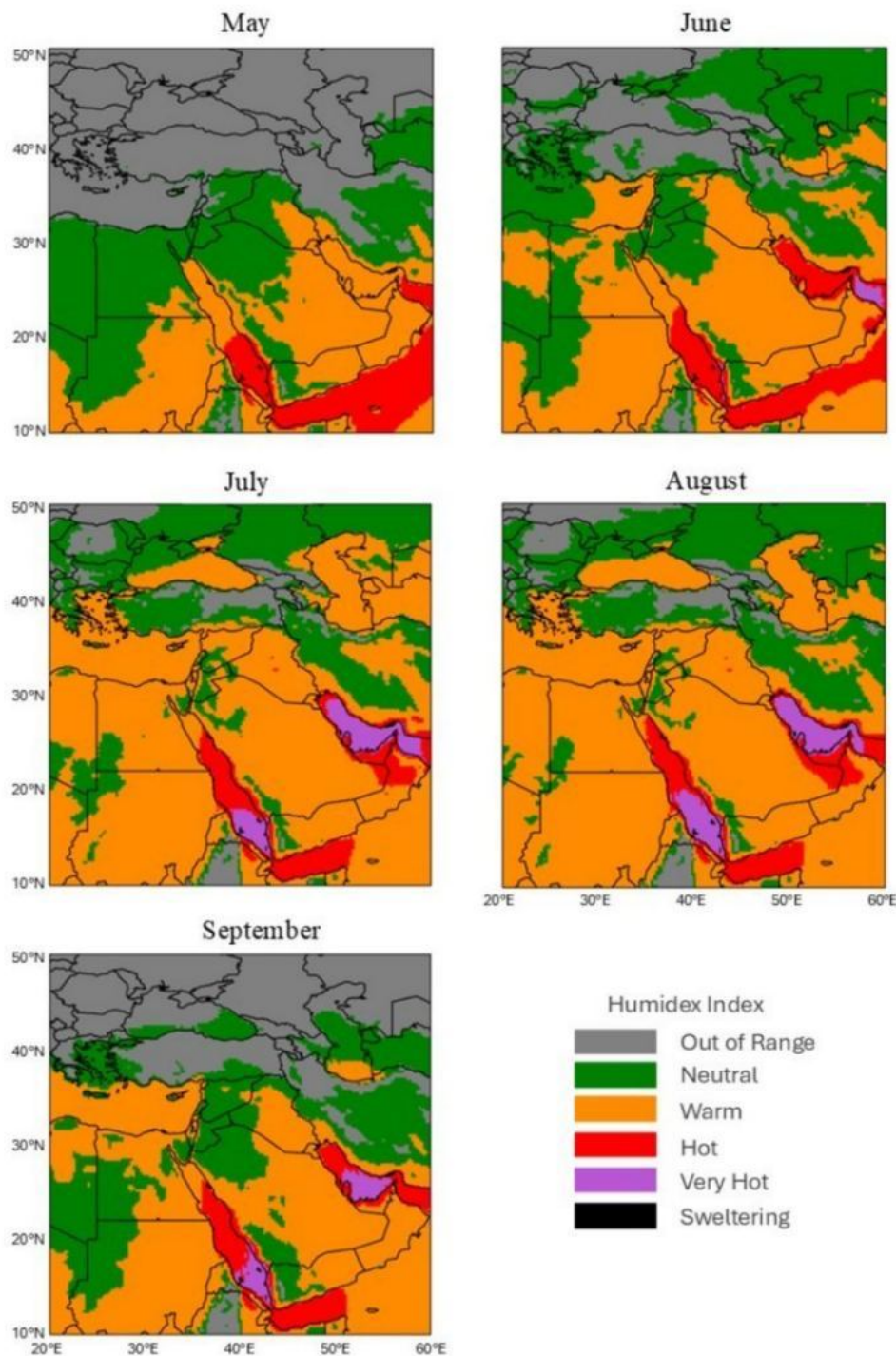


FIGURE 10 | Colour maps showing the 1980–2023 average of the HD index in the study region for May to September.

August. In contrast, all indices show strong agreement in identifying neutral thermal sensations in most northern cities, particularly during the transitional months of May and September.

Identifying shifts in thermal comfort indices is essential for understanding how climate change is altering human heat exposure across the EMME region. Figure 15 compares the first (1980–1989) and last (2014–2023) decades and highlights region-specific shifts in thermal sensation. These shifts reflect changes in key climate variables, particularly T , RH and

WS, and expose vulnerabilities in both arid inland areas and humid coastal regions. For instance, humidity-sensitive indices such as WBGT detect a notable shift to hotter conditions over water bodies such as the Black and Caspian Seas. In contrast, indices that depend more strongly on wind speed, such as ET, indicate worsening heat stress over high-wind desert interiors. Together, these patterns illustrate how multi-index shift analysis provides a more nuanced characterisation of evolving thermal risk and can support locally tailored planning and adaptation.

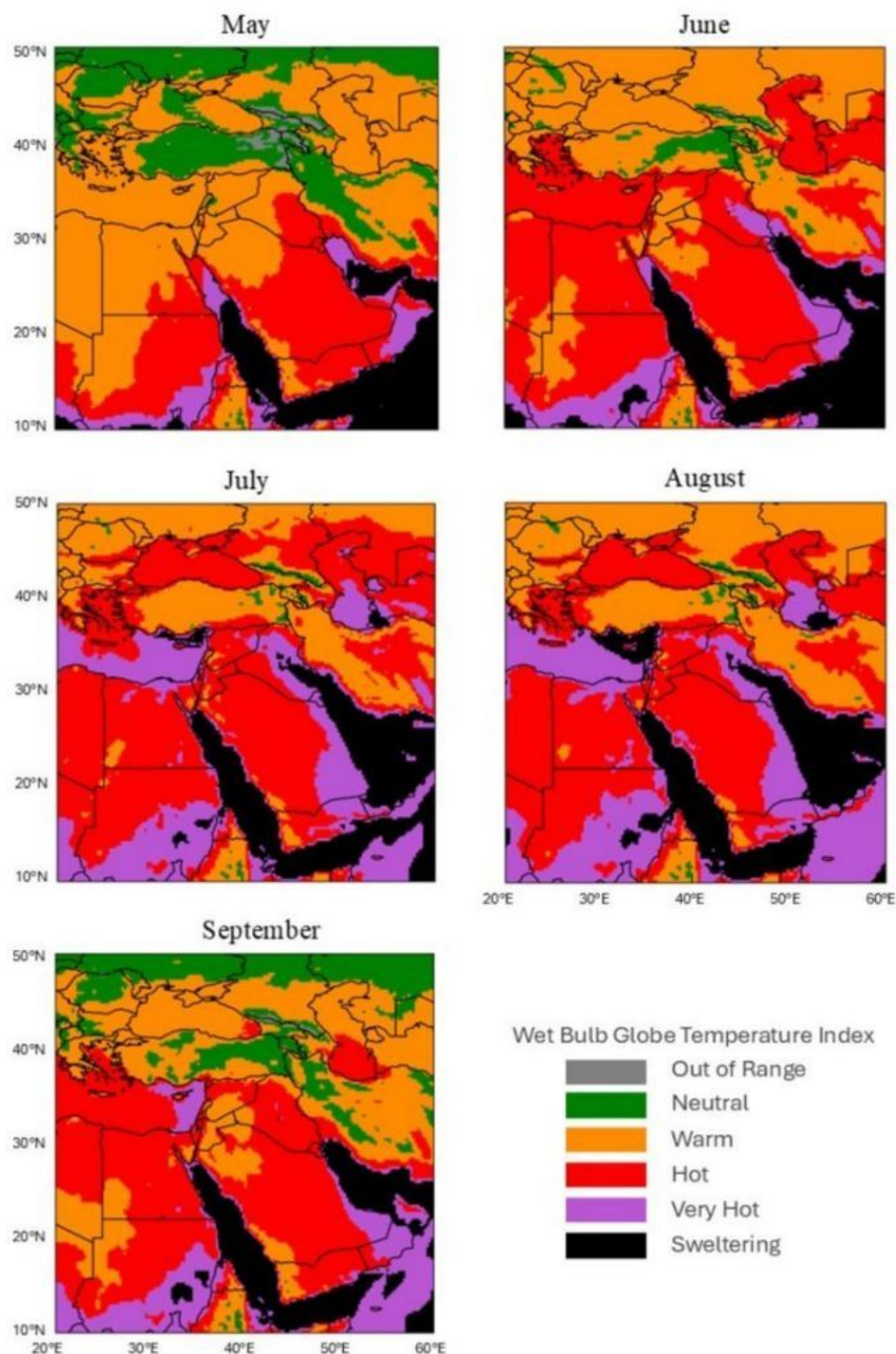


FIGURE 11 | Colour maps showing the 1980–2023 average of the WBGT index in the study region for May to September.

To further quantify outdoor thermal comfort in major cities across the study region, we computed the average number of hours per day during summer associated with hot, very hot and sweltering sensations. These results are presented in Figure 16 for all indices considered. In each subfigure, the angle represents the year, from 1980 (year 1) to 2023 (year 44). For each year, the number of daily discomfort hours is shown; points farther from the centre indicate higher discomfort. Each line colour corresponds to a different index, as indicated in the legend. For all cities, the mean number of uncomfortable hours

per day increases over time across indices, indicating a clear warming trend. For example, in Doha (DI), the mean number of uncomfortable hours per day ranges from 12.50 to 20.43 over 1980–2000 and from 15.98 to 18.44 over 2000–2023. Similarly, in Amman (ET), the mean number of uncomfortable hours per day ranges from 1.59 to 4.75 over 1980–2000 and increases to 3.25 to 6.25 over 2000–2023. Across cities, indices show consistent temporal patterns, with WBGT predicting the highest discomfort hours, followed by ET, HI, DI and HD. For example, in Cairo, WBGT predicts 16.37 to 22.14 uncomfortable hours per day over

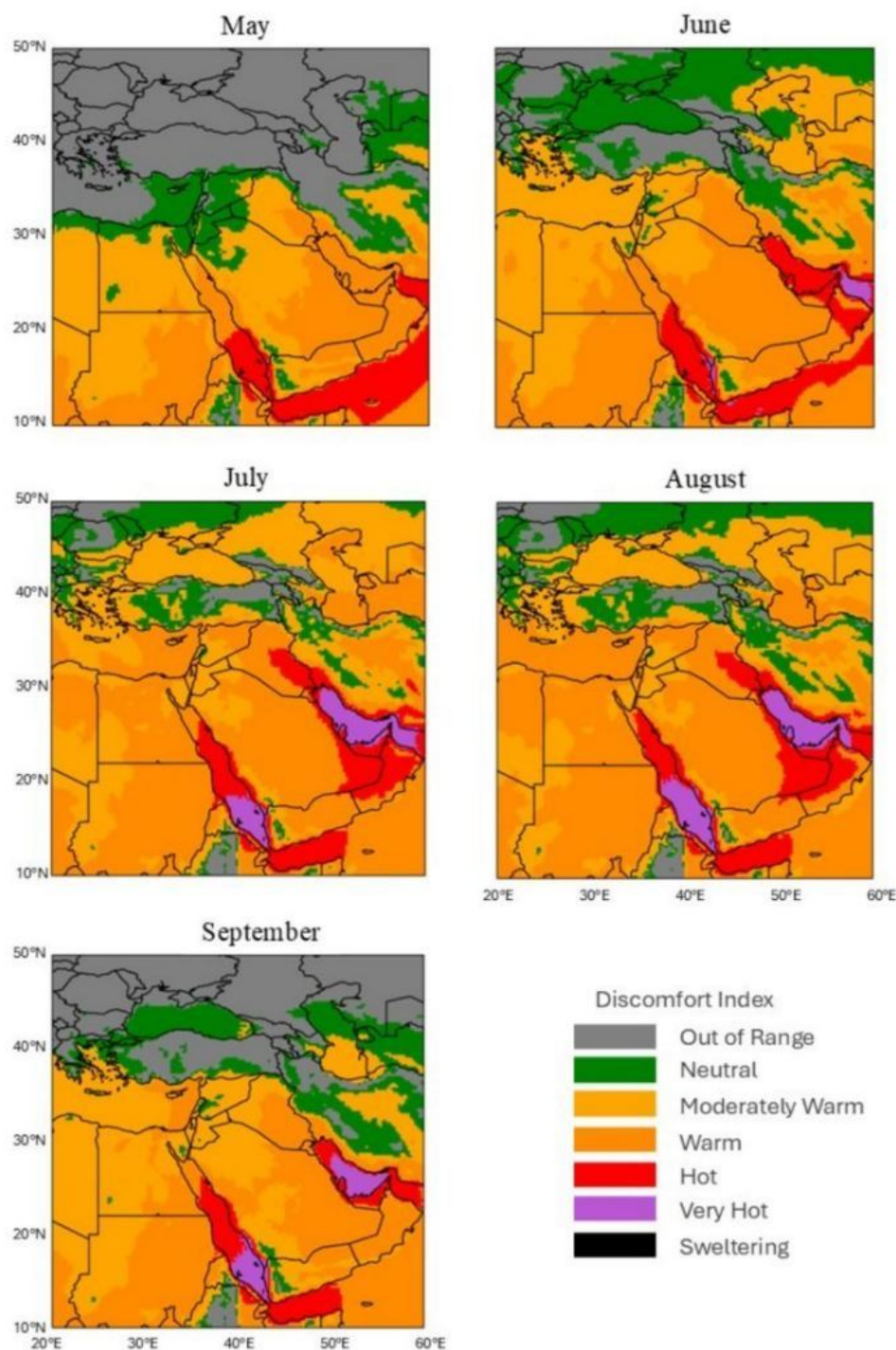


FIGURE 12 | Colour maps showing the 1980–2023 average of DI in the study region for May to September.

the study period. The corresponding ranges for ET, HI, DI and HD are 6.46 to 11.43, 3.08 to 8.01, 0.68 to 4.99 and 0.09 to 3.48, respectively.

Figure 17 shows standardised anomalies of the mean number of summer days per year during which ET indicates hot, very hot or sweltering sensations. The reference baseline is the mean number of discomfort days over 1980–2023. ET is selected because it depends on T , RH and WS, thereby incorporating both convective and evaporative heat-exchange

mechanisms. Bars are colour-coded to indicate El Niño, La Niña or neutral phases, enabling assessment of the influence of large-scale teleconnections on discomfort levels. Particular attention is given to the 1997–1998 El Niño event, widely regarded as a benchmark mega (or super) El Niño episode due to its exceptional intensity. This event has been linked to changes in large-scale circulation (Hansen et al. 2012) and has been discussed in recent studies focused on the Arabian Peninsula (AlSarmi and Washington 2011; Dasari et al. 2022; Saharwardi et al. 2023). Furthermore, 1997–1998 was identified as one

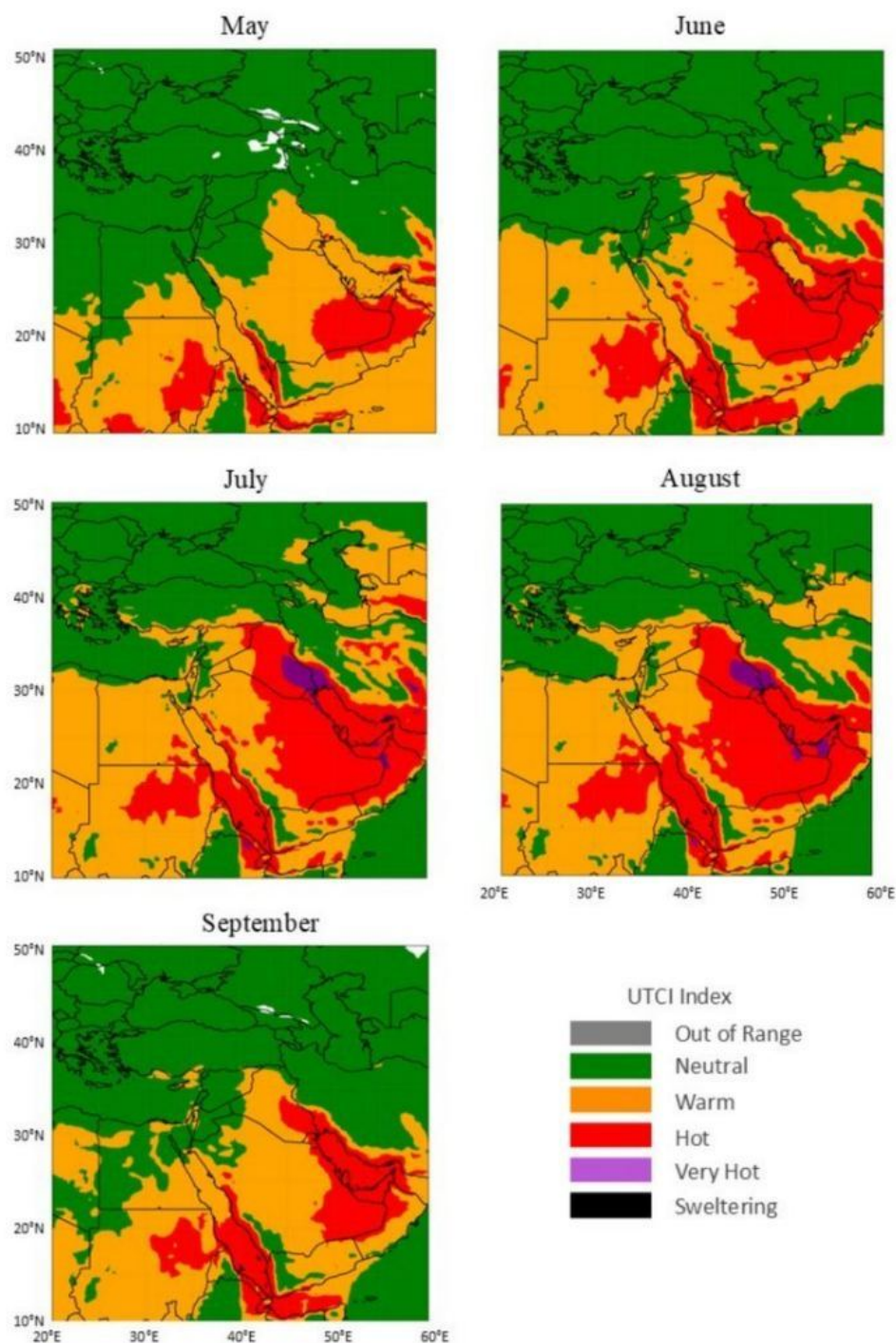


FIGURE 13 | Colour maps showing the 1980–2023 average of UTCI in the study region for May to September.

of the driest periods in the Eastern Mediterranean region (Oroud 2018; Giorgi and Lionello 2008). Two additional mega El Niño episodes have been recorded since the beginning of reliable global climate observations: 1982–1983 and 2015–2016 (Abhik et al. 2021). Such episodes can cause global-scale disruptions, including droughts, floods, coral bleaching and impacts on agriculture and ecosystems.

The results indicate a transition from predominantly negative to consistently positive anomalies around 1998, suggesting a shift towards above-average summer discomfort days after this

period. This transition is consistent with concurrent changes in T and RH anomalies (Figure S.1 in Supporting Information). Sana'a is not included in Figure 17 because it does not exhibit summer discomfort days, consistent with Figure 16. To objectively identify change points in the anomaly time series, we applied Pettitt's test, a non-parametric method used to detect shifts in median values. Table 6 indicates early change points for Nicosia (1993) and for Baghdad, Kuwait City, Manama and Riyadh (1995), followed by a broad clustering around 1997 for most remaining cities; later change points occur for Tehran (2004) and for Cairo and Muscat (2005).

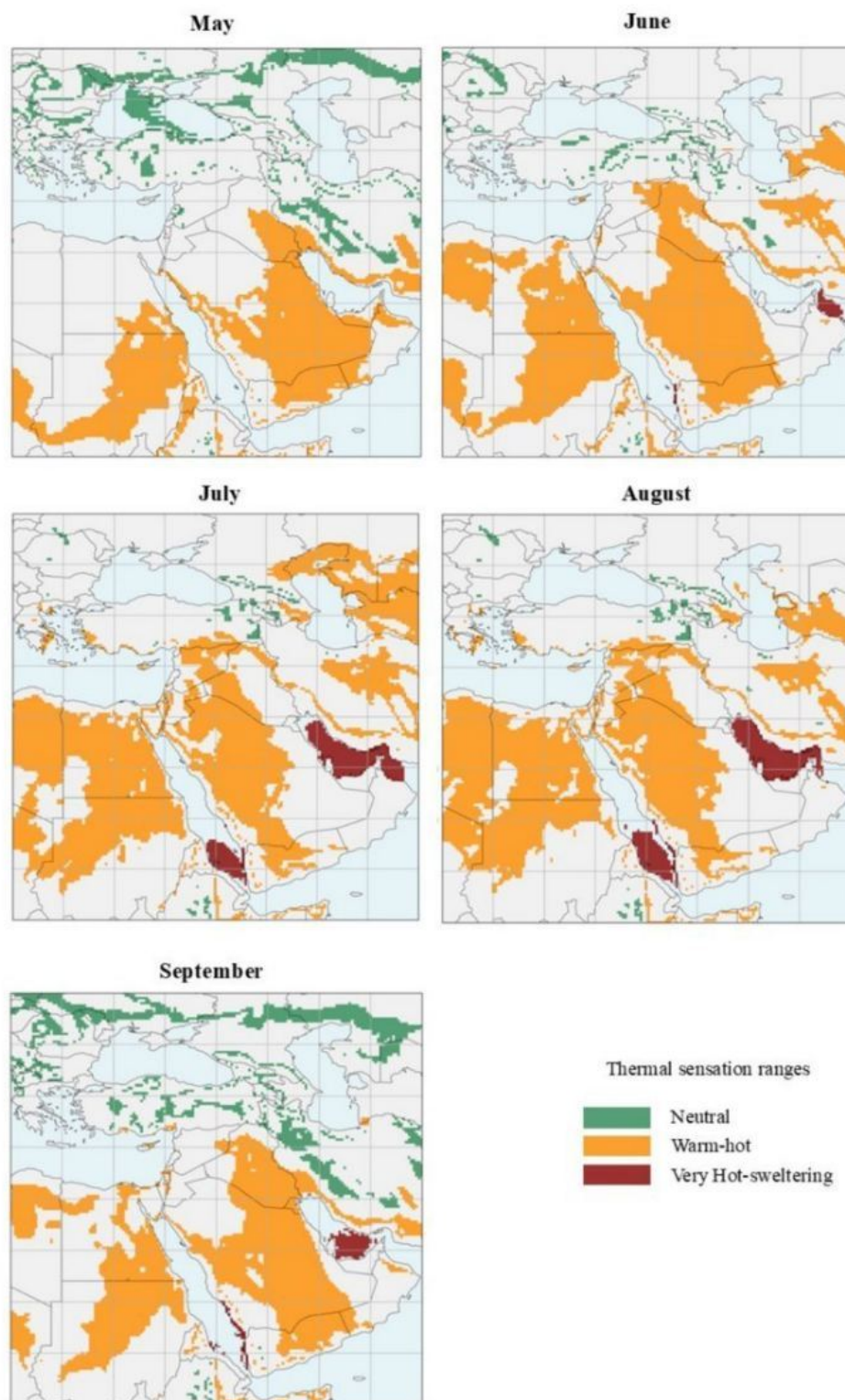


FIGURE 14 | Maps showing the spatial zones where all studied indices (ET, HI, HD, WBGT and DI) indicate the same thermal sensation category for each summer month, averaged over 1980–2023. Thermal sensation is grouped into neutral, warm-hot and very hot-sweltering categories.

Inspection of La Niña occurrences shows that their frequency increased after 1998, with 10 occurrences during 1999–2023 (40%) compared to 5 occurrences during 1980–1998 (26.32%), as shown in Figure 17. Multiple studies have shown that the La Niña phase of ENSO can trigger widespread drought and

heat stress across the EMME (Donat et al. 2014; S. Wang et al. 2014; R. Kumar et al. 2018). This behaviour has been linked to Pacific Ocean cooling that can intensify the Indian monsoon, redirecting moisture towards South Asia and away from the Middle East and thereby promoting drought

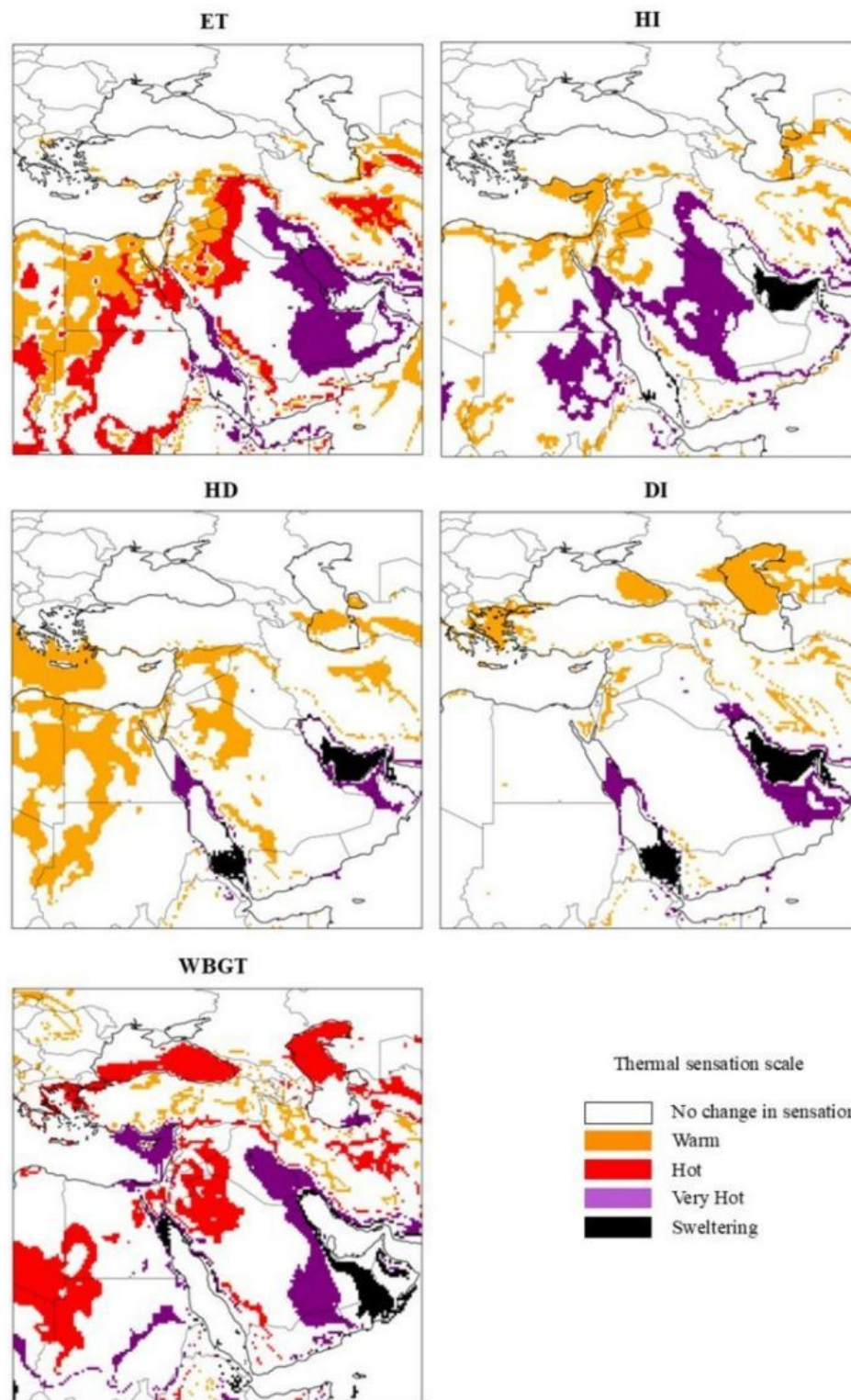


FIGURE 15 | Spatial shifts in thermal discomfort categories based on different thermal comfort indices between the first decade (1980–1989) and the last decade (2014–2023) of the study period across the EMME region.

conditions and exacerbated heatwaves. Hoell et al. (2024) document an increase in the frequency and severity of compound drought–heat events across the Middle East and Southwest Asia since 1999 and relate this shift to persistent La Niña-like sea surface temperature patterns that induce subsidence and warming. Similarly, Rezaei (2023) identifies robust statistical

associations between La Niña phases and increased drought frequency across arid and semi-arid zones of the Middle East, particularly over the eastern Mediterranean, Iraq and Iran. Dasari et al. (2017) highlight rainfall suppression over the Red Sea convergence zone during La Niña years due to weakened moisture convergence, affecting western Saudi Arabia

and Yemen. Mahmoud et al. (2023) report that La Niña conditions lead to prolonged droughts and decreased river flows in key watersheds, with projected amplification under climate change scenarios. From a paleoclimate perspective, Kushnir

and Stein (2019) show that La Niña-like conditions during the Mediaeval Warm Period were historically linked to reduced precipitation and enhanced aridity across the Levant. Additionally, La Niña events are associated with reduced RH,

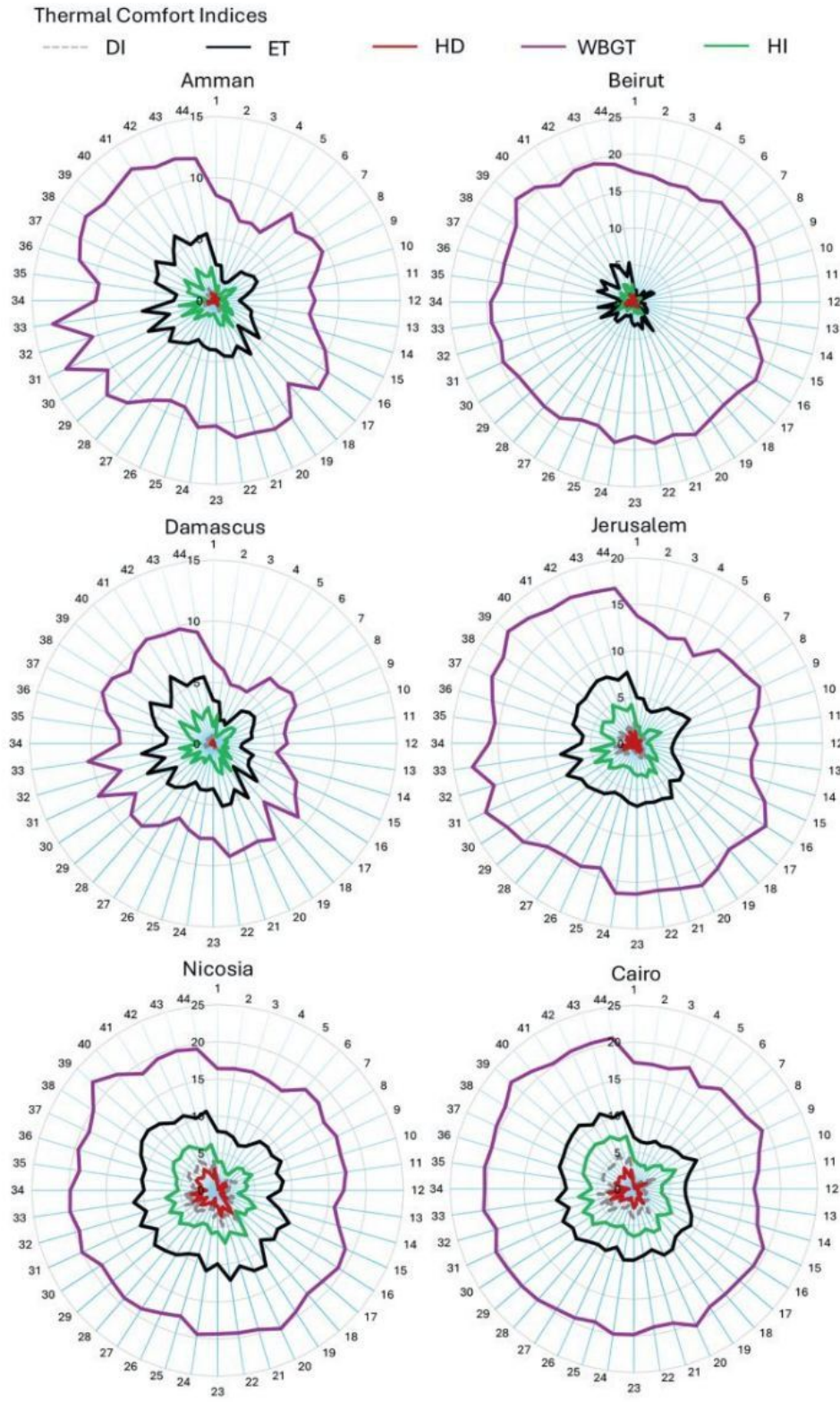


FIGURE 16 | Annual trend of the average number of discomfort hours per day during summer in the central (Amman, Beirut, Damascus, Jerusalem and Nicosia) and southwestern (Cairo) cities. Annual trend of the average number of discomfort hours per day during summer in the northern (Ankara, Athens, Istanbul and Tehran) and southeastern (Abu Dhabi and Baghdad) cities.

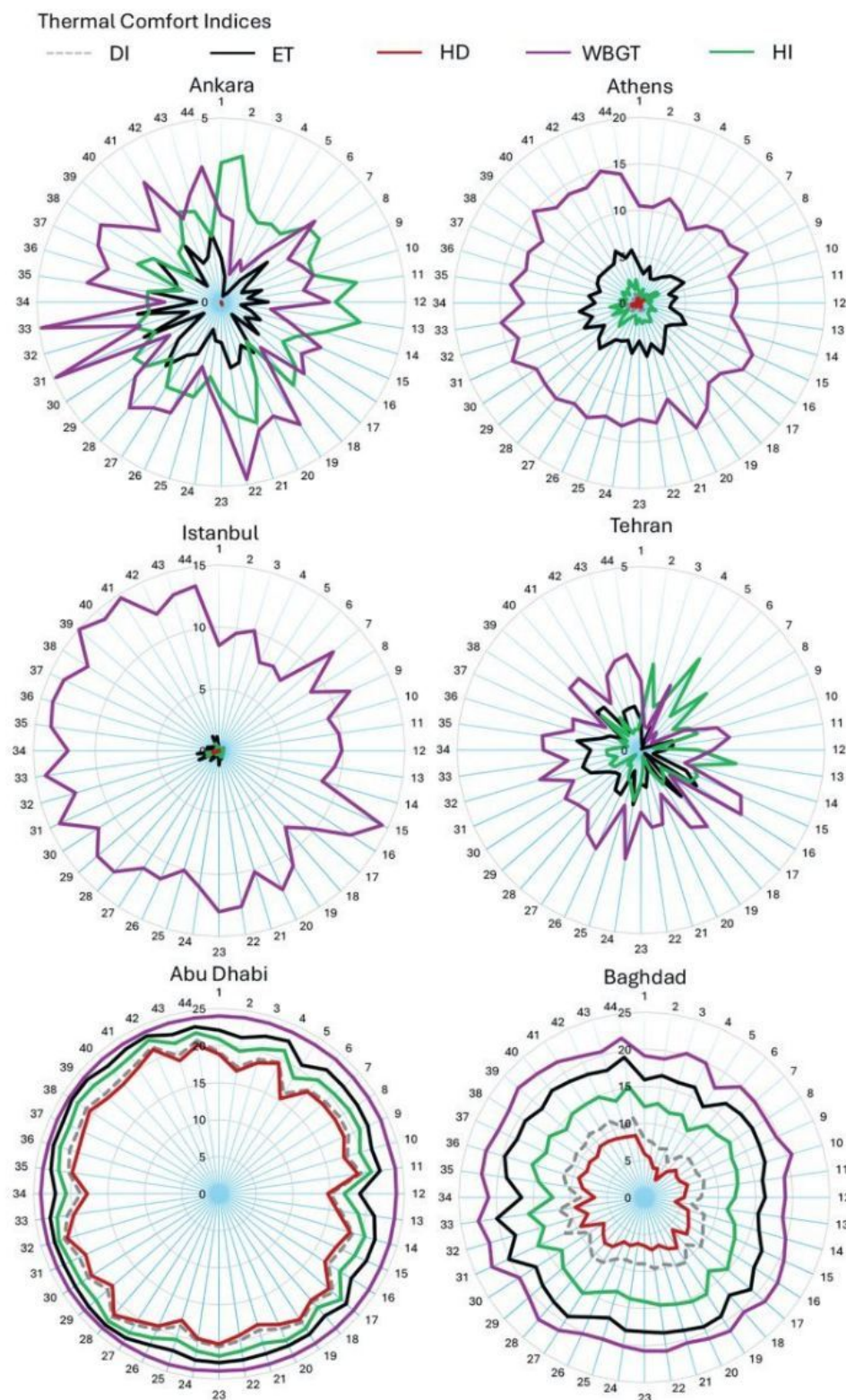


FIGURE 16 | (Continued)

which can further amplify heat stress and thermal discomfort (Adhikari and Wang 2025).

Analysis of the linear trends in Figure 17 reveals an increasing number of discomfort days over 1980–2023 for all cities. For central, southwestern and northern cities, trends exhibit positive slopes both before and after 1998, indicating a continuous

increase in discomfort days. For southeastern cities, trends show a negative slope in phase 1 (1980–1998) and a nearly zero slope in phase 2 (1999–2023). Similar patterns are observed for standardised anomalies of the mean annual number of summer days for which WBGT indicates hot, very hot, or sweltering sensations; results are presented in Figure S.3 of the [Supporting Information](#).

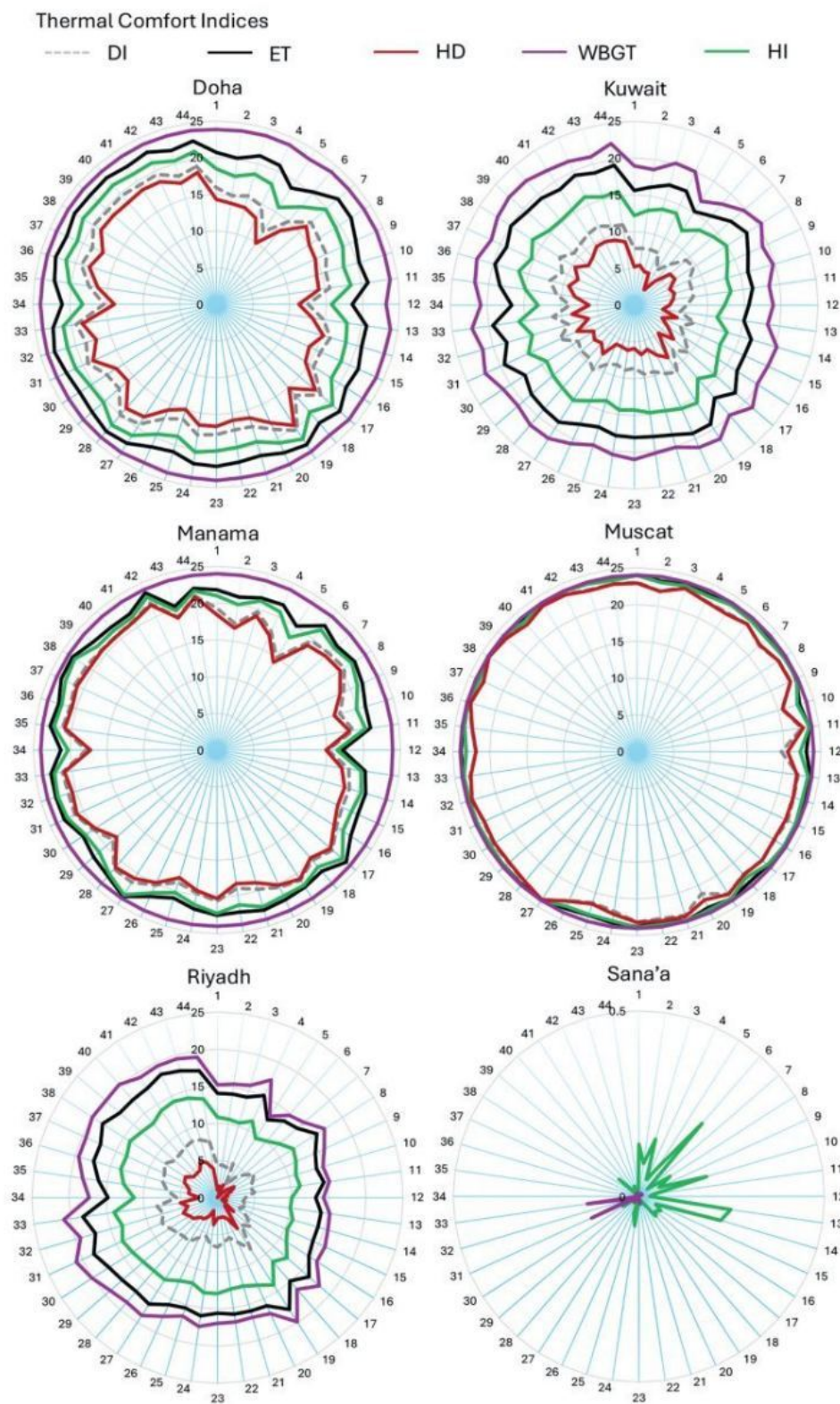
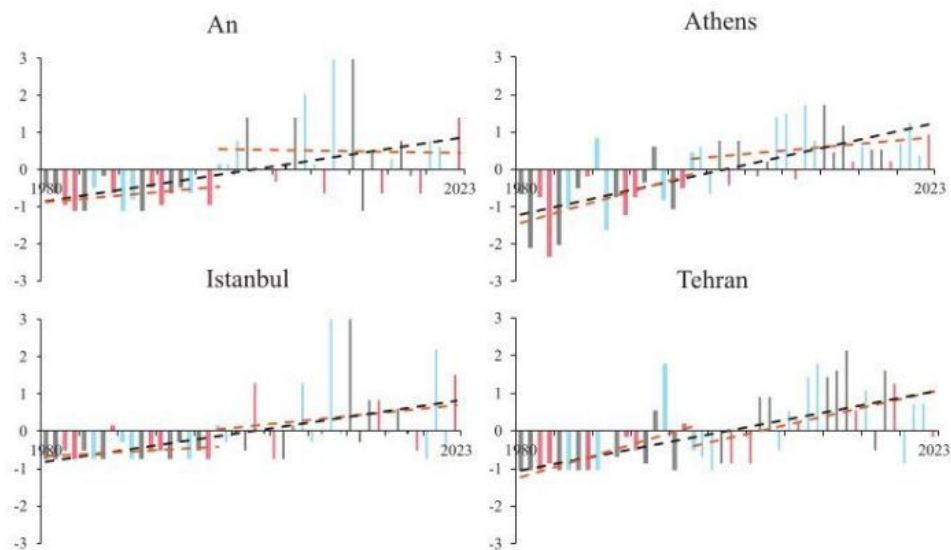


FIGURE 16 | (Continued)

Figure 18 shows standardised anomalies of the mean number of summer days per year in which ET indicates hot, very hot, or sweltering sensations across the study region. Bars are colour-coded according to positive, negative, or neutral IOD phases. Compared to ENSO, IOD exhibits fewer negative events and more positive events after 1998. Negative IOD events decrease from 36.84% during 1980–1998 to 4.00% during 1999–2023,

while positive events increase from 10.53% to 16.00% over the same periods. A prolonged period of neutral IOD conditions is also observed from 1998 until 2014. Previous studies have shown that positive IOD phases are associated with warmer sea surface temperatures that enhance moisture transport from the Arabian Sea, modify monsoonal winds and shift precipitation patterns, contributing to heat stress in the EMME region



(Raicich et al. 2003; Hochman and Gildor 2025; Abdul Muttalib et al. 2020). Fonseca et al. (2022) report a significant influence of positive IOD events on atmospheric circulation over the Arabian Peninsula. During these phases, above-average sea surface temperatures in the western Indian Ocean enhance evaporation, increasing atmospheric moisture that is advected

inland and contributing to a northward displacement of inter-tropical discontinuities over the Arabian Peninsula. Chandran et al. (2016) show that precipitation and temperature variability over the UAE is closely associated with large-scale forcings and that the positive IOD phase is linked to increased rainfall over the region.

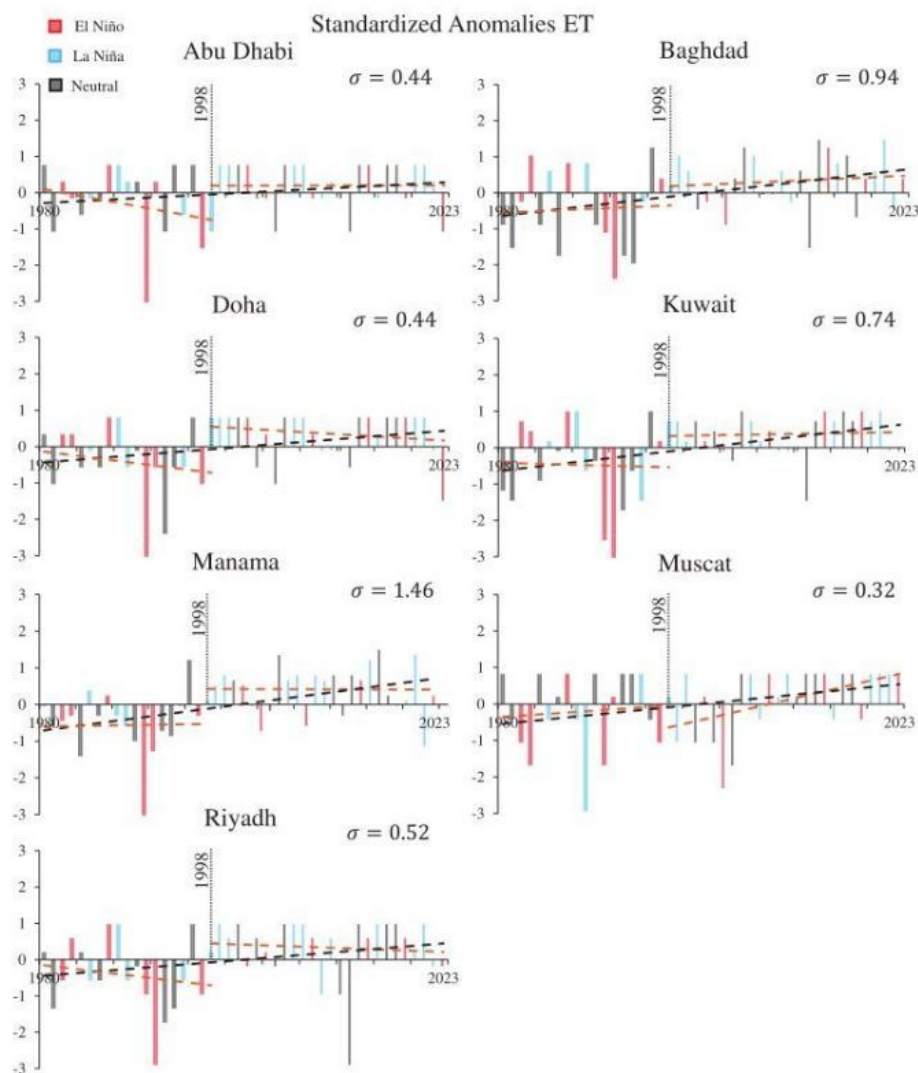


FIGURE 17 | (Continued)

Several studies have also examined the interaction between ENSO and IOD. Ashok et al. (2003) show that some strong positive IOD events are not associated with ENSO, suggesting that the Indian Ocean can sustain strong IOD events without external ENSO forcing (e.g., 1994; Figures 17 and 18). Yuan and Li (2008) investigate decadal variability in the IOD–ENSO relationship and conclude that IOD is not driven solely by ENSO, but can actively influence ENSO intensity. Increased ENSO strength has also been linked to more frequent positive IOD events (Annamalai et al. 2005). Strong El Niño events often coincide with positive IOD phases, as observed in 1997, 2015, 2019 and 2023 (Figures 17 and 18). Chandran et al. (2016) further conclude that, depending on phase and amplitude, IOD events can either amplify or dampen ENSO-driven precipitation anomalies over the UAE.

To illustrate the climatic shift after 1998, the middle column of Figure 19 presents differences in mean summer T , RH and WS between phase 2 (1999–2023) and phase 1 (1980–1998). The maps show that warming not only persists but intensifies across the EMME during phase 2. Average summer T increases by 0.53°C, 0.40°C, 0.56°C and 0.51°C in the central, southeastern, southwestern and northern city groups, respectively, from phase 1 to phase 2. Similarly, drying intensifies during phase 2: RH

declines by 0.42%, 0.19% and 0.66% in the central, southwestern and northern city groups, respectively, while a slight increase of 0.098% is observed in the southeastern group. Wind-speed changes also vary spatially: compared to phase 1, stronger WS occurs in the central and northern regions during phase 2, while the southeastern and southwestern regions experience calmer conditions.

To highlight the intensification of the meteorological conditions affecting thermal comfort, the right column of Figure 19 presents differences in mean summer T , RH and WS between phase 2 (1999–2023) and the full study period (1980–2023). Summer T during phase 2 exceeds the long-term mean across all regions, confirming its dominant contribution to regional warming. For RH, phase 2 values are lower than the long-term mean in most northern and central cities, suggesting a shift towards more arid conditions (e.g., Tehran: 27.19% in phase 2 vs. 28.75% over 1980–2023; Jerusalem: 53.78% vs. 54.96%). For WS, positive anomalies occur in the central and northern regions, while negative anomalies occur in the southeastern and southwestern regions. Together, these results underscore the characterisation of the post-1998 era as a dominant and transformative phase in the regional climate trajectory.

TABLE 6 | Year of the anomaly shift predicted using Pettitt's test for each city over 1980–2023 and the corresponding statistical *p*-value.

Cities	Year of anomaly shift	Statistical <i>p</i> -value
Nicosia	1993	0.000031
Baghdad	1995	0.03668
Kuwait City	1995	0.034
Manama	1995	0.00127
Riyadh	1995	0.055
Amman	1997	0.000017
Ankara	1997	0.000112
Athens	1997	0.0000699
Beirut	1997	0.00000664
Damascus	1997	0.0000106
Doha	1997	0.04
Istanbul	1997	0.001
Jerusalem	1997	0.000000937
Abu Dhabi	1998	0.49
Tehran	2004	0.000101
Cairo	2005	0.0000112
Muscat	2005	0.03786

Overall, our results indicate that increasingly frequent La Niña phases are associated with intensified summer thermal discomfort across the EMME region, consistent with previous studies. ENSO phases modulate large-scale atmospheric circulation patterns that influence *T*, RH, WS, precipitation and other climatic variables in the region. In particular, La Niña is associated with changes in the Walker circulation and jet stream trajectories that can suppress rainfall in some regions, enhance subsidence and heat buildup over the Middle East, and disrupt monsoon dynamics.

Annual trend of the average number of discomfort hours per day during summer in southeastern cities (Doha, Kuwait City, Manama, Muscat, Riyadh and Sana'a).

4 | Discussion and Concluding Remarks

This study provides a comprehensive assessment of the spatial and temporal evolution of thermal discomfort across the EMME using ERA5 reanalysis over 43 years (1980–2023). By analysing key meteorological variables (*T*, RH and WS) and applying five thermal comfort indices (ET, HI, HD, WBGT and DI), we characterise interannual variability, diurnal behaviour and regional disparities. We also examine the influence of large-scale climate variability, particularly ENSO and IOD phases.

We further assess thermal sensation using the physiology-based UTCI. Differences between UTCI and HI are minor and are

largely attributable to limitations of HI, notably the absence of a neutral thermal sensation category, and to discrepancies over water bodies where thermal sensation is of limited practical relevance (primarily marine traffic). Overall, incorporating human physiological parameters through UTCI does not substantially alter regional-scale thermal comfort patterns compared with meteorological indices alone, supporting the use of simpler indices (e.g., HI) for characterising summer thermal discomfort in the EMME within the scope of this study.

The results demonstrate a clear warming trend across all examined cities, with the most pronounced summer temperature increases occurring in the northern and central regions. In contrast, RH trends exhibit spatial variability: central and northern cities show consistent declines, while parts of the southeastern and southwestern coastal zones exhibit moderate increases. These RH patterns are shaped by atmospheric circulation dynamics, including large-scale subsidence, which can suppress humidity inland while enhancing surface evaporation over adjacent water bodies.

Across the EMME region, both daytime and nighttime temperatures have increased, with nighttime warming particularly pronounced. This asymmetric warming contributes to prolonged thermal stress, especially in urbanised areas where heat retention can be amplified by anthropogenic influences. Thermal indices confirm heightened stress during peak summer months (July–August), particularly in southeastern cities, while northern and some central cities experience more moderate—though still increasing—levels of discomfort.

All thermal comfort indices examined indicate rising discomfort trends. Central cities show the steepest increases in the number of discomfort hours per day, whereas southeastern cities start from higher baseline discomfort levels and exhibit comparatively more stable long-term trends. This apparent plateauing may be linked to moderating influences from nearby water bodies (e.g., the Red Sea and Arabian Gulf) and/or prevailing wind regimes. The indices also exhibit different sensitivities to meteorological drivers: WBGT responds strongly to *T* (and is less sensitive to RH), ET is more strongly modulated by WS, and HI and HD indicate higher discomfort under high-RH conditions.

A significant climatic transition is evident around 1998, associated with the strong 1997–1998 El Niño and a positive IOD phase. In the subsequent period, La Niña phases became more frequent and positive IOD events increased. These changes have been linked to enhanced compound drought and heat events across the region and are reflected in the thermal comfort trends, with evidence of intensified and more frequent extreme conditions after 1998. Although the rate of increase in discomfort levels slows slightly after 1998 in some cities, the cumulative increase remains substantial.

The spatial distribution of thermal sensation shifts also highlights the influence of geography and atmospheric dynamics. WBGT, which incorporates humidity effects, indicates a shift towards hotter categories over major water bodies such as the Black and Caspian Seas and along the Arabian Gulf and Mediterranean coastlines. ET, which is less sensitive to RH and more sensitive to WS, indicates increasing thermal stress over

arid inland regions, consistent with the role of wind in modulating perceived heat. HI, HD and DI show a general shift towards warmer categories across both coastal and inland zones. A multi-index comparison over 1980–2023 further reveals consistent regional classification: the southeastern region—notably Riyadh, Baghdad and Kuwait City—is persistently

characterised as warm to hot during summer, while very hot to sweltering conditions are identified in Doha, Manama and Abu Dhabi. Cairo is classified as warm to hot during July and August. In contrast, northern cities are consistently classified as thermally neutral during the transitional months of May and September.

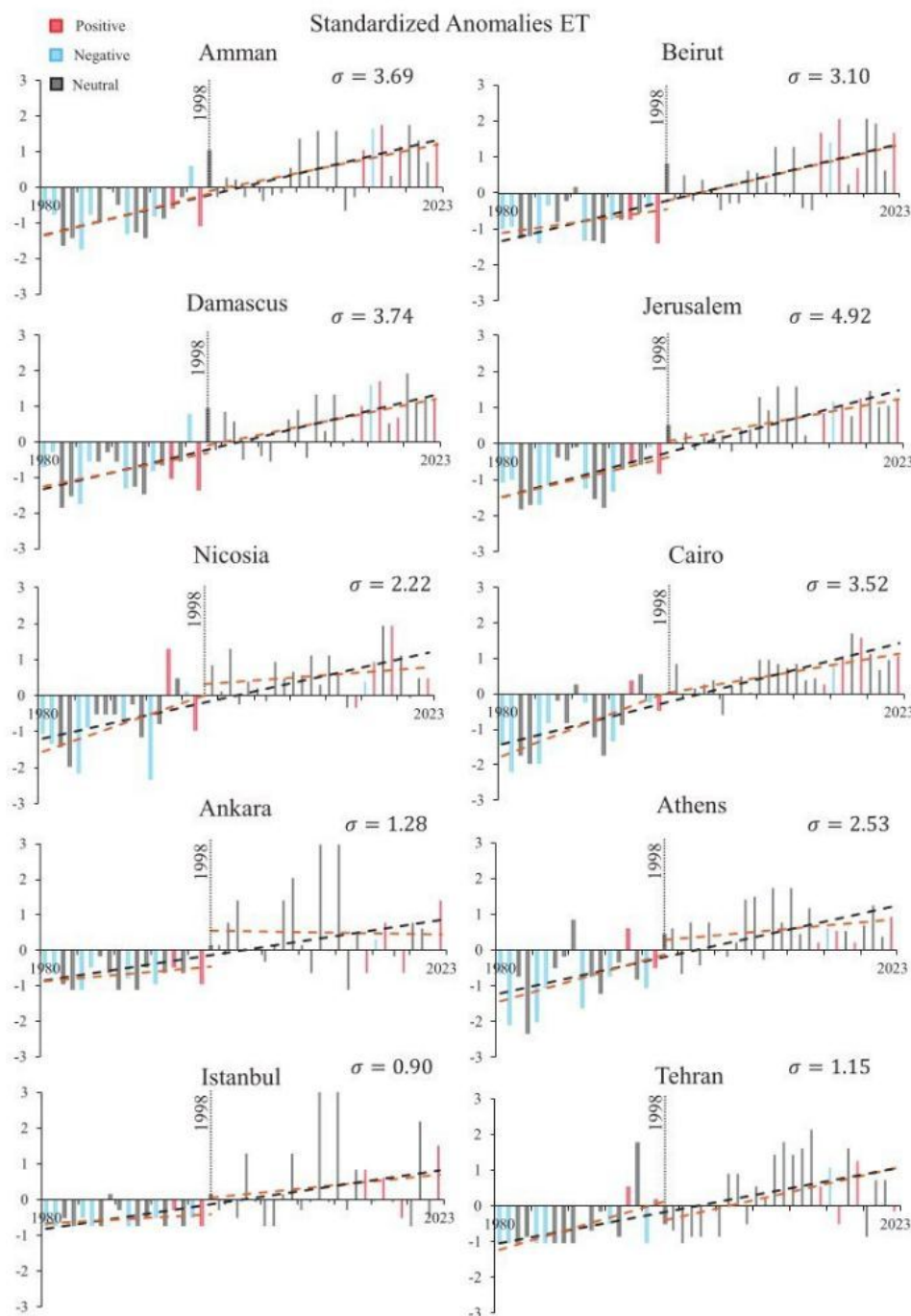


FIGURE 18 | Standardised anomalies of the average number of summer days during which ET indicates hot, very hot or sweltering sensations for more than eight consecutive hours in Amman, Beirut, Damascus, Jerusalem, Nicosia, Cairo, Ankara, Athens, Istanbul and Tehran over 1980–2023. Positive, negative and neutral IOD events are shown by red, blue and black bars, respectively. The black dotted line shows the linear trend over 1980–2023, and the orange dotted lines show the linear trends for phases 1 and 2 separately. Standardised anomalies of the average number of summer days during which ET indicates hot, very hot, or sweltering sensations for more than eight consecutive hours in Abu Dhabi, Baghdad, Doha, Kuwait City, Manama, Muscat and Riyadh over 1980–2023. Positive, negative and neutral IOD events are shown by red, blue and black bars, respectively. The black dotted line shows the linear trend over 1980–2023, and the orange dotted lines show the linear trends for phases 1 and 2 separately.

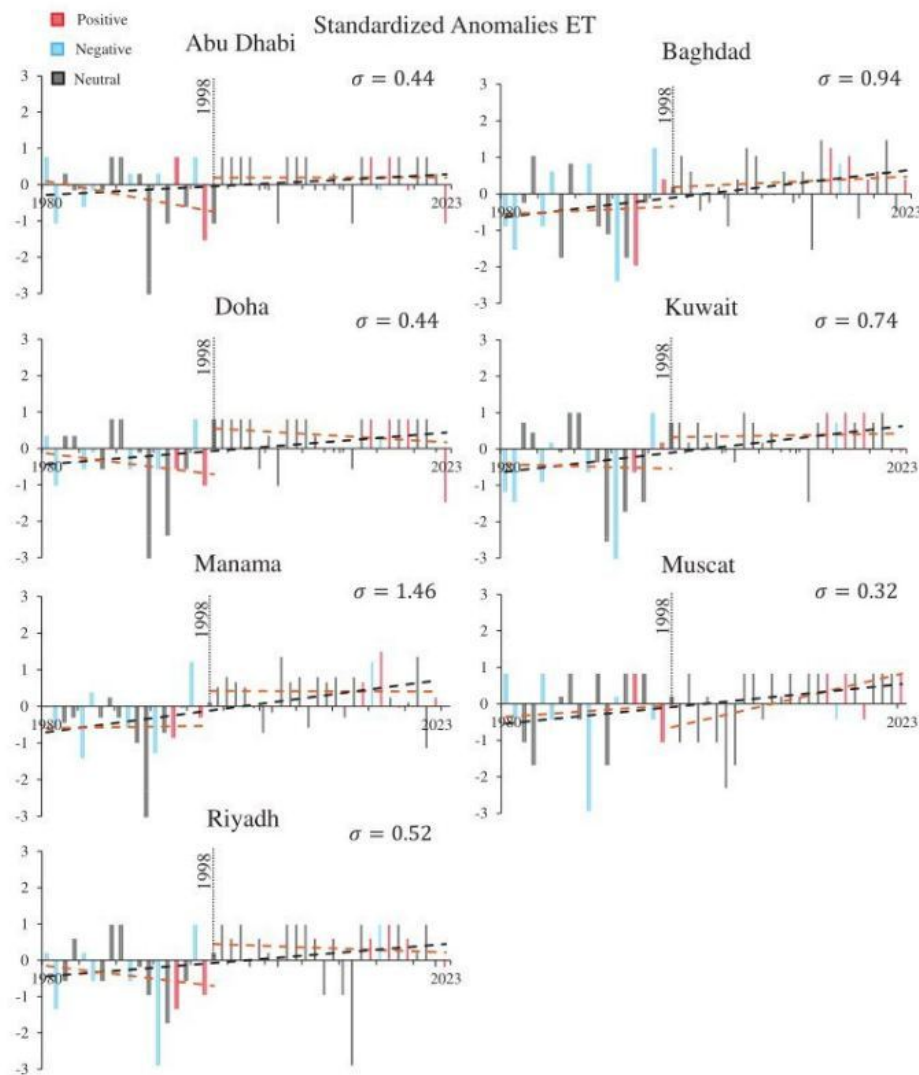


FIGURE 18 | (Continued)

Regional wind patterns, especially the Shamal winds, are important for interpreting these variations. These dry, north-westerly winds commonly affect the Arabian Peninsula and can reduce RH and promote evaporative cooling. However, they may also increase perceived dryness and dehydration risk, and their net effect depends on temperature and humidity conditions. The interaction between WS and T in ET highlights this dual influence. The ET contours in Figure 7 indicate that at lower air temperatures, higher wind speeds enhance convective heat loss and are associated with colder perceived sensations. In contrast, when air temperature exceeds skin temperature (approximately 37°C), higher wind speeds can exacerbate thermal stress by increasing convective heat gain, potentially offsetting evaporative cooling benefits.

The region's surrounding seas—the Mediterranean to the west; the Black and Caspian Seas to the north; the Arabian Gulf to the east; and the Red Sea and Indian Ocean to the south—exert considerable influence on local and regional climate. These act as reservoirs of heat and moisture, drive sea-breeze circulations and modulate air–sea interactions. Coupled ocean–atmosphere phenomena such as ENSO and IOD further affect precipitation,

cloud cover and circulation patterns that ultimately influence thermal comfort.

This study also identifies regions of unambiguous heat stress, where all five indices converge on the same thermal category. In particular, southeastern cities (e.g., Riyadh and Kuwait City) are consistently classified as warm to hot throughout summer, while northern cities remain thermally neutral in May and September. In addition, comparing the first (1980–1989) and last (2014–2023) decades reveals clear shifts in thermal sensation categories across the region: indices sensitive to RH show stronger warming over water bodies, while wind-sensitive indices capture increasing stress inland. These consistent multi-index patterns increase confidence in the results and support their translation into planning and policy across sectors such as urban design, public health and agriculture.

In conclusion, rising thermal discomfort across the EMME reflects a complex interplay of global climate signals, regional atmospheric dynamics and geographic feedbacks linked to land–sea contrasts. The results underscore the importance of considering not only temperature changes but also RH, WS and

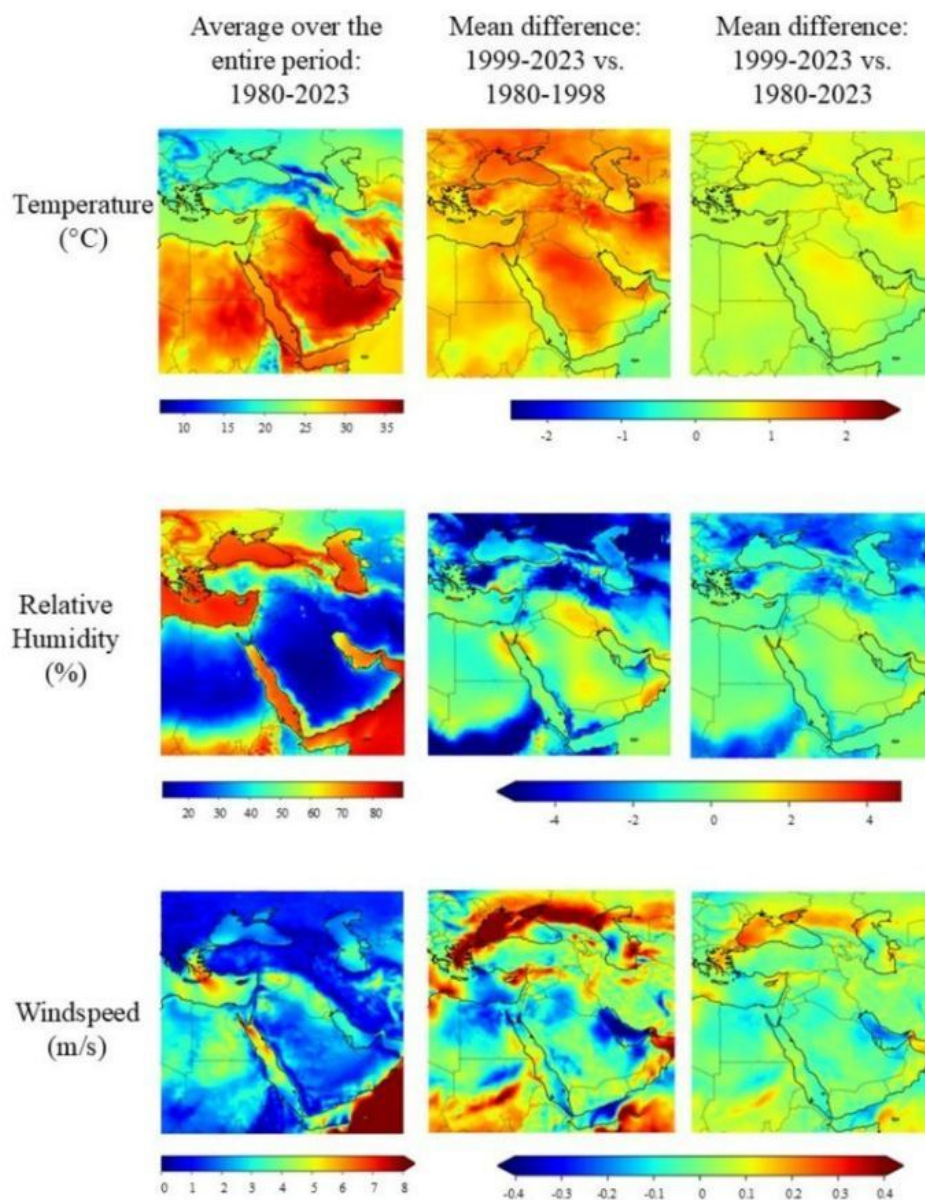


FIGURE 19 | Maps showing summer (May–September) mean patterns of temperature ($^{\circ}\text{C}$), relative humidity (%) and wind speed (m s^{-1}). Column 1 shows climatological means over 1980–2023. Column 2 shows differences between phase 2 and phase 1 means. Column 3 shows differences between phase 2 and the full-period (1980–2023) means.

their interactions when evaluating human thermal discomfort. By providing a region-wide, long-term assessment using multiple indices, this work can inform climate-based early warning for heat stress, support the design of targeted heat action plans and public health interventions and guide urban and regional planning in areas where future development may intensify heat-related risks. Finally, the multi-index framework presented here offers a quantitative basis for comparing thermal discomfort across regions and time periods, supporting climate-adaptation planning, occupational health considerations and resilience-oriented policy development.

5 | Limitations and Future Work

An important limitation of this study is the use of ERA5 reanalysis data, which does not explicitly resolve urban-scale thermal

processes (e.g., UHI effects). As a result, thermal discomfort in densely built environments may be underestimated. A recent study of the Urban Heat Island (UHI) effect in large Pakistani cities (Khan and Shahid 2024) shows that rapid urbanisation, increased impervious surfaces, reduced vegetation, anthropogenic heat emissions and urban morphology can substantially elevate near-surface air temperatures, particularly at night, thereby prolonging and intensifying heat stress. These findings are relevant to rapidly growing cities across the EMME region and suggest that ERA5-based discomfort indices likely provide conservative estimates in densely built areas. Accordingly, while the present analysis robustly captures regional-scale variability and long-term trends, it likely underrepresents city-scale heat exposure. Future work should integrate high-resolution urban observations, satellite-derived land surface temperature and/or urban climate models to better account for UHI effects and refine urban heat-risk assessments.

Author Contributions

Jennifer Karam: conceptualization, methodology, formal analysis, visualization, writing original draft; **Nareg Karaoghlanian:** methodology, data curation, visualization, formal analysis; **Omar Hakla:** data curation, formal analysis; **Louis Youssef:** data curation, formal analysis; **Jad Bhamdouni:** data curation, formal analysis; **Hari Prasad Dasari:** conceptualization, writing-review and editing, methodology, supervision; **Ibrahim Hoteit:** conceptualization, writing-review and editing, methodology, resources; **Issam Lakkis:** conceptualization, methodology, writing-review and editing, supervision, resources, funding acquisition.

Acknowledgements

This research was supported by the AUB Research Board (URB) and the Maroun Semaan Faculty of Engineering and Architecture at the American University of Beirut.

Funding

This work was supported by the University Research Board and American University of Beirut.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data presented in this work, along with the software used for analysis, are deposited in the open access GitHub repository <https://github.com/JennyferKaram/Thermal-comfort-data>.

References

- Abdul Muttalib, A. M., S. M. Ameen, and A. B. Mahmood. 2020. "The Impacts of the Pacific Southern Oscillation and the Indian Ocean Dipole on the Mean Sea Level of the Arabian Gulf and the Arabian Sea." *Journal of King Abdulaziz University – Marine Sciences* 30, no. 2: 17–31.
- Abhik, S., P. Hope, H. H. Hendon, et al. 2021. "Influence of the 2015–2016 El Niño on the Record-Breaking Mangrove Dieback Along Northern Australia Coast." *Scientific Reports* 11, no. 1: 20411.
- Adhikari, G. P., and G. Wang. 2025. "Understanding the Driving Mechanism for Heat Waves Variability Over South Asia During El Niño and La Niña Events." *Atmospheric Research* 321: 108076.
- Aghamolaei, R., and A. Lak. 2023. "Outdoor Thermal Comfort for Active Ageing in Urban Open Spaces: Reviewing the Concepts and Parameters." *Ageing International* 48, no. 2: 438–451.
- Al-Sakkaf, A. S., Z. Jiahua, Y. Fengmei, et al. 2024. "Assessing Exposure to Climate Extremes Over the Arabian Peninsula Using ERA5 Reanalysis Data: Spatial Distribution and Temporal Trends." *Atmospheric Research* 300: 107224.
- AlSarmi, S., and R. Washington. 2011. "Recent Observed Climate Change Over the Arabian Peninsula." *Journal of Geophysical Research: Atmospheres* 116, no. D11: D11109.
- Annamalai, H., S. Xie, J. McCreary, and R. Murtugudde. 2005. "Impact of Indian Ocean Sea Surface Temperature on Developing El Niño." *Journal of Climate* 18, no. 2: 302–319.
- Aragão, L., and F. Porcù. 2022. "Cyclonic Activity in the Mediterranean Region From a High-Resolution Perspective Using ECMWF ERA5 Dataset." *Climate Dynamics* 58, no. 5: 1293–1310.
- Ashok, K., Z. Guan, and T. Yamagata. 2003. "A Look at the Relationship Between the ENSO and the Indian Ocean Dipole." *Journal of the Meteorological Society of Japan. Series II* 81, no. 1: 41–56.
- Barbariol, F., S. Davison, F. M. Falcieri, et al. 2021. "Wind Waves in the Mediterranean Sea: An ERA5 Reanalysis Wind-Based Climatology." *Frontiers in Marine Science* 8: 760614.
- Chandran, A., G. Basha, and T. Ouarda. 2016. "Influence of Climate Oscillations on Temperature and Precipitation Over The United Arab Emirates." *International Journal of Climatology* 36, no. 1: 225–235.
- Coccolo, S., J. Kämpf, J. L. Scartezzini, and D. Pearlmutter. 2016. "Outdoor Human Comfort and Thermal Stress: A Comprehensive Review on Models and Standards." *Urban Climate* 18: 33–57.
- Cook, E. R., R. Seager, Y. Kushnir, et al. 2015. "Old World Megadroughts and Pluvials During the Common Era." *Science Advances* 1, no. 10: e1500561.
- Copernicus Climate Change Service. 2025. *Global Climate Highlight Report 2024*. European Center for Medium-Range Weather Forecasts (ECMWF).
- Dasari, H. P., S. Desamsetti, S. Langodan, Y. Viswanadhapalli, and I. Hoteit. 2021. "Analysis of Outdoor Thermal Discomfort Over the Kingdom of Saudi Arabia." *GeoHealth* 5, no. 6: e2020GH000370.
- Dasari, H. P., S. Langodan, and Y. Viswanadhapalli. 2017. "ENSO Influence on the Interannual Variability of the Red Sea Convergence Zone and Associated Rainfall. KAUST Repository." <https://repository.kaust.edu.sa/handle/10754/625880>.
- Dasari, H. P., Y. Viswanadhapalli, S. Langodan, et al. 2022. "High-Resolution Climate Characteristics of the Arabian Gulf Based on a Validated Regional Reanalysis." *Meteorological Applications* 29, no. 5: e2102.
- de Freitas, C. R., and E. A. Grigorieva. 2017. "A Comparison and Appraisal of a Comprehensive Range of Human Thermal Climate Indices." *International Journal of Biometeorology* 61: 487–512.
- Donat, M., T. Peterson, M. Brunet, et al. 2014. "Changes in Extreme Temperature and Precipitation in the Arab Region: Long-Term Trends and Variability Related to ENSO and NAO." *International Journal of Climatology* 34, no. 3: 581.
- Du, C., B. Li, W. Yu, H. Liu, C. Li, and R. Yao. 2019. "Moisture in Clothing and Its Transient Influence on Human Thermal Responses Through Clothing Microenvironment in Cold Environments in Winter." *Building and Environment* 150: 1–12.
- Elnabawi, M. H., and N. Hamza. 2019. "Behavioural Perspectives of Outdoor Thermal Comfort in Urban Areas: A Critical Review." *Atmosphere* 11, no. 1: 51.
- Fonseca, R., D. Francis, N. Nelli, and M. Thota. 2022. "Climatology of the Heat Low and the Intertropical Discontinuity in the Arabian Peninsula." *International Journal of Climatology* 42, no. 2: 1092–1117.
- Gandham, H., H. P. Dasari, T. M. Luong, et al. 2025. "Declining Summer Circulation Over the Eastern Mediterranean and Middle East." *npj Climate and Atmospheric Science* 8, no. 1: 1–11.
- Gauthier, S. "Developing a Method to Monitor Thermal Discomfort Response Variability." (PhD thesis, University College London, 2015).
- Giorgi, F., and P. Lionello. 2008. "Climate Change Projections for the Mediterranean Region." *Global and Planetary Change* 63, no. 2–3: 90–104.
- Hamzeh, N. H., S. Karami, D. G. Kaskaoutis, I. Tegen, M. Moradi, and C. Opp. 2021. "Atmospheric Dynamics and Numerical Simulations of Six Frontal Dust Storms in the Middle East Region." *Atmosphere* 12, no. 1: 125.
- Hansen, J., R. Ruedy, M. Sato, and K. Lo. 2012. "Global Temperature in 2011, Trends, and Prospects. GISS Surface Temperature Analysis."
- Hersbach, H., B. Bell, P. Berrisford, G. Biavati, A. Horányi, and J. M. Sabater. 2023. "ERA5 Hourly Data on Pressure Levels From 1940 to Present: Tech. Rep. Copernicus Climate Change Service (C3S), Climate Data Store (CDS)."

- Hochman, A., and H. Gildor. 2025. "Synergistic Effects of El Niño–Southern Oscillation and the Indian Ocean Dipole on Middle Eastern Subseasonal Precipitation Variability and Predictability." *Quarterly Journal of the Royal Meteorological Society* 151, no. 766: e4903.
- Hoell, A., R. Robinson, L. Agel, et al. 2024. "Changes to Middle East and Southwest Asia Compound Drought and Heat Since 1999." *Journal of Climate* 37, no. 1: 269–287.
- Huang, C. H., H. H. Tsai, and H. C. Chen. 2020. "Influence of Weather Factors on Thermal Comfort in Subtropical Urban Environments." *Sustainability* 12, no. 5: 2001.
- Ioannou, L. G., J. Foster, N. B. Morris, et al. 2022. "Occupational Heat Strain in Outdoor Workers: A Comprehensive Review and Meta-Analysis." *Temperature* 9, no. 1: 67–102.
- Jendritzky, G., R. De Dear, and G. Havenith. 2012. "UTCI—Why Another Thermal Index?" *International Journal of Biometeorology* 56, no. 3: 421–428.
- Kaynakli, O., M. Mutlu, I. Atmaca, and M. Kilic. 2014. "Investigation of Humidity Effects on the Thermal Comfort and Heat Balance of the Body." In *Progress in Exergy, Energy, and the Environment*, edited by I. Dincer, A. Midilli, and H. Kucuk, 421–434. Springer International Publishing.
- Khan, N., S. Ahmad, and S. Shahid. 2025. "Assessing the Deadly Heat Waves Over South Asian Cities: An Insights From UTCI and WBGT Indices." *International Journal of Climatology* 45, no. 7: e8806.
- Khan, N., Z. Ismail, M. I. Asiri, et al. 2025. "Escalating Heat Stress and Outdoor Labor Productivity Loss in the Arabian Peninsula: A Five-Decade Analysis of Climate Change Impacts." *Earth Systems and Environment* 9: 1–25.
- Khan, N., M. Kamruzzaman, and S. Shahid. 2025. "Diurnal Pattern of Heat Stress Over South Asia: A Wet Bulb Globe Temperature-Based Analysis From 1984 to 2023." *International Journal of Climatology* 45, no. 6: e8797.
- Khan, N., and S. Shahid. 2024. "Urban Heat Island Effect and Its Drivers in Large Cities of Pakistan." *Theoretical and Applied Climatology* 155, no. 6: 5433–5452.
- Kumar, P., and A. Sharma. 2020. "Study on Importance, Procedure, and Scope of Outdoor Thermal Comfort – A Review." *Sustainable Cities and Society* 61: 102297. <https://www.sciencedirect.com/science/article/pii/S2210670720305187>.
- Kumar, R., R. Attada, H. P. Dasari, et al. 2018. "Aerosol Optical Depth Variability Over the Arabian Peninsula as Inferred From Satellite Measurements." *Atmospheric Environment* 187: 346–357.
- Kushnir, Y., and M. Stein. 2019. "Medieval Climate in the Eastern Mediterranean: Instability and Evidence of Solar Forcing." *Atmosphere* 10, no. 1: 29. <https://www.mdpi.com/2073-4433/10/1/29/pdf>.
- Lally, V. E., and B. F. Watson. 1960. "Humiture Revisited." *Weatherwise* 13, no. 6: 254–256.
- Lukić, M. 2019. "An Analysis of the Influence of Air Temperature and Humidity on Outdoor Thermal Comfort in Belgrade (Serbia) Using a Simple Heat Index." *Archives for Technical Sciences* 21: 75–84.
- Mahmoud, S. H., T. Y. Gan, and D. Z. Zhu. 2023. "Impacts of Climate Change and ENSO on Water Resources and Drought in an Arid Region." *Climate Dynamics* 61: 4079. <https://link.springer.com/article/10.1007/s00382-023-06795-7>.
- Missenard, F. A. 1933. *Etude physiologique et technique de la ventilation*. Librairie De L'Enseignement Technique.
- Neelin, J., C. Chou, and H. Su. 2003. "Tropical Drought Regions in Global Warming and El Niño Teleconnections." *Geophysical Research Letters* 30, no. 24: 2275.
- Nelli, N., D. Francis, A. Alkatheeri, and R. Fonseca. 2024. "Evaluation of Reanalysis and Satellite Products Against Ground-Based Observations in a Desert Environment." *Remote Sensing* 16, no. 19: 3593.
- NOAA (National Oceanic and Atmospheric Administration). 2024. "Climate Prediction Center (CPC)." <https://origin.cpc.ncep.noaa.gov/>.
- Olano Pozo, J. X., Ò. Saladié, and A. Boqué-Ciurana. 2024. "Rising Temperatures, Wavering Human Towers? Temperature Trends and Thermal Comfort During Castells Exhibitions in Catalonia (1951–2023). Case Studies in Valls (24 June), La Bisbal del Penedès (15 August), Tarragona (19 August), and Vilafranca del Penedès (30 August)." *Climate* 12, no. 8: 112.
- Oroud, I. M. 2018. "Global Warming and Its Implications on Meteorological and Hydrological Drought in the Southeastern Mediterranean." *Environmental Processes* 5, no. 2: 329–348.
- Panagopoulos, T., J. A. G. Duque, and M. B. Dan. 2016. "Urban Planning With Respect to Environmental Quality and Human Well-Being." *Environmental Pollution* 208: 137–144.
- Patel, R., J. King, L. Phelps, and D. Sanderson. 2017. *What Practices Are Used to Identify and Prioritize Vulnerable Populations Affected by Urban Humanitarian Emergencies? A Systematic Review*. Humanitarian Evidence Programme. Feinstein International Center.
- Polonsky, A. B., and D. V. Basharin. 2017. "How Strong Is the Impact of the Indo-Ocean Dipole on the Surface Air Temperature/Sea Level Pressure Anomalies in the Mediterranean Region?" *Global and Planetary Change* 151: 101–107.
- Potchter, O., P. Cohen, T. P. Lin, and A. Matzarakis. 2018. "Outdoor Human Thermal Perception in Various Climates: A Comprehensive Review of Approaches, Methods and Quantification." *Science of the Total Environment* 631: 390–406.
- Ragatoa, D. S., L. K. Amekudzi, A. H. Fink, et al. 2024. "Droughts and Heatwaves in the West African Monsoon System." *International Journal of Climatology* 44, no. 11: 3681–3705.
- Raich, F., N. Pinardi, and A. Navarra. 2003. "Teleconnections Between Indian Monsoon and Sahel Rainfall and the Mediterranean." *International Journal of Climatology* 23, no. 2: 173–186.
- Rezaei, A. 2023. "Teleconnections Between Ocean–Atmosphere Circulations and Historical Integrated Drought in the Middle East and North Africa." *Environmental Monitoring and Assessment* 195: 775. <https://doi.org/10.1007/s10661-023-11386-4>.
- Rijal, H. B. 2012. "Thermal Adaptation Outdoors and the Effect of Wind on Thermal Comfort." In *Ventilating Cities: Air-Flow Criteria for Healthy and Comfortable Urban Living*, edited by S. Kato and K. Hiyama, 33–58. Springer Netherlands.
- Saharwardi, M. S., H. P. Dasari, V. Aggarwal, K. Ashok, and I. Hoteit. 2023. "Long-Term Variability in the Arabian Peninsula Droughts Driven by the Atlantic Multidecadal Oscillation." *Earth's Future* 11, no. 11: e2023EF003549.
- Saji, N., and T. Yamagata. 2003. "Possible Impacts of Indian Ocean Dipole Mode Events on Global Climate." *Climate Research* 25, no. 2: 151–169.
- Santee, W. R., and R. F. Wallace. 2005. "Comparison of Weather Service Heat Indices Using a Thermal Model." *Journal of Thermal Biology* 30, no. 1: 65–72.
- Schott, F. A., S. P. Xie, and J. P. McCreary Jr. 2009. "Indian Ocean Circulation and Climate Variability." *Reviews of Geophysics* 47, no. 1: RG1002.
- Shimazaki, Y., A. Yoshida, and T. Yamamoto. 2015. "Thermal Responses and Perceptions Under Distinct Ambient Temperature and Wind Conditions." *Journal of Thermal Biology* 49: 1–8.
- Simplemaps, 2023. "World Cities Database." <https://simplemaps.com/data/world-cities/>.
- Steadman, R. G. 1979. "The Assessment of Sultriness. Part I: A Temperature-Humidity Index Based on Human Physiology and Clothing Science." *Journal of Applied Meteorology* 18, no. 7: 861–873.

- Thom, E. C. 1959. "The Discomfort Index." *Weatherwise* 12, no. 2: 57–61.
- Ullah, S., A. Aldossary, W. Ullah, and S. G. Al-Ghamdi. 2024. "Augmented Human Thermal Discomfort in Urban Centers of the Arabian Peninsula." *Scientific Reports* 14, no. 1: 3974.
- Ullah, W., K. Alabduoli, S. Ullah, et al. 2024. "Comparison of 2-m Surface Temperature Data Between Reanalysis and Observations Over the Arabian Peninsula." *Atmospheric Research* 311: 107725.
- Varlas, G., K. Stefanidis, G. Papaioannou, et al. 2022. "Unravelling Precipitation Trends in Greece Since 1950s Using ERA5 Climate Reanalysis Data." *Climate* 10, no. 2: 12.
- Wang, G. 2005. "Agricultural Drought in a Future Climate: Results From 15 Global Climate Models Participating in the IPCC 4th Assessment." *Climate Dynamics* 25: 739–753.
- Wang, S., J. Huang, Y. He, and Y. Guan. 2014. "Combined Effects of the Pacific Decadal Oscillation and El Nino-Southern Oscillation on Global Land Dry–Wet Changes." *Scientific Reports* 4, no. 1: 6651.
- Wolter, K., and M. S. Timlin. 1998. "Measuring the Strength of ENSO Events: How Does 1997/98 Rank?" *Weather* 53, no. 9: 315–324.
- Wong, M. C., J. Wang, X. Zhi, and L. Dong. 2024. "A 1940–2020 Spatiotemporal Analysis of Thermal Discomfort Days in Southeast Asian Countries." *Environmental Research Communications* 6, no. 10: 101009.
- Wu, J., Z. Hu, Z. Han, Y. Gu, L. Yang, and B. Sun. 2021. "Human Physiological Responses of Exposure to Extremely Cold Environments." *Journal of Thermal Biology* 98: 102933.
- Yaglou, C., and D. Minaed. 1957. "Control of Heat Casualties at Military Training Centers." *Archives of Industrial Health* 16, no. 4: 302–316.
- Yang, J., W. Weng, and B. Zhang. 2014. "Experimental and Numerical Study of Physiological Responses in Hot Environments." *Journal of Thermal Biology* 45: 54–61.
- Yilmaz, M. 2023. "Accuracy Assessment of Temperature Trends From ERA5 and ERA5-Land." *Science of the Total Environment* 856: 159182.
- Yuan, Y., and C. Li. 2008. "Decadal Variability of the IOD-ENSO Relationship." *Chinese Science Bulletin* 53, no. 11: 1745–1752.
- Zittis, G., M. Almazroui, P. Alpert, et al. 2022. "Climate Change and Weather Extremes in the Eastern Mediterranean and Middle East." *Reviews of Geophysics* 60, no. 3: e2021RG000762.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information
Data S2: Supporting Information